Relaxing Entry Constraints: College Expansion and Tertiary Attainment in Rural India*

Ravinder†

October, 2026

Abstract

Tertiary education attainment among young adults in India has risen sharply over the past three decades, alongside a large expansion in the number of higher education institutions. This paper asks whether expanded access to colleges raised local rural tertiary education attainment. I study Rajasthan, which follows a similar trend to India in rural tertiary attainment but has experienced the highest rate of college expansion among Indian states. I combine geocoded college locations from the All India Survey on Higher Education with unique cohort-level records from the Socio-Economic and Caste Census 2011. Using individuals aged 22 to 31 in 2011, I construct a village-by-birth-cohort panel spanning 1998 to 2007, linking each cohort to the college supply prevailing when it reached college-entry age. The expansion began in 2003, and I exploit the staggered entry of colleges across villages. Exposure to a college within 15 km raised the tertiary attainment of exposed cohorts by about 1.4 percentage points two years after exposure and by 2.2-2.4 percentage points after four to five years, with no evidence of pre-trends. The effect is robust to several checks, including alternative comparison groups, distance thresholds, restricted cohort window, and adjustments for local economic activity, and holds for both men and women. Turning to college placement, district land fractionalization in 1999 is strongly associated with subsequent college growth. More- and less-fractionalized districts had similar college stocks before the expansion but diverged only after it began. This pattern is consistent with fragmented, lower-value agricultural land offering cheaper sites for a land-intensive, largely private expansion of higher education.

Keywords: tertiary education institution expansion; educational attainment; land fractionalization

JEL Classification: I23, I25, I28, R52

*This is the third chapter of my PhD thesis. I am grateful to my advisor, Prof. Abhiroop Mukhopadhyay, for his valuable guidance throughout this paper. I am also thankful to him for generously sharing the Rajasthan cohort-specific, village-level SECC data on tertiary education completion. All remaining errors are my own.

†Ph.D. Candidate, Indian Statistical Institute, Delhi. Email: ravinder20r@isid.ac.in

1 Introduction

Tertiary education is a key driver of productivity and economic growth. Across countries, college graduates raise the productivity of those who work alongside them (Moretti, 2004a); regions with more universities grow faster (Valero and Van Reenen, 2019); and private returns to higher education have risen over recent decades (Psacharopoulos and Patrinos, 2018). Yet these gains are not distributed evenly across space. The returns to skill are highest in cities, where dense labor markets, agglomeration economies, and concentrations of skilled workers reward graduates most (Glaeser and Maré, 2001; Moretti, 2004b). Colleges, particularly those established by private, fee-financed entrants, therefore tend to locate where prospective students and the returns to education are concentrated: in and around towns.

Evidence from India tells a similar story. Growth across Indian states responds more strongly to the share of the population completing tertiary education than to reductions in illiteracy (Castelló-Climent and Mukhopadhyay, 2013); higher education raises district-level economic development (Castelló-Climent et al., 2018); and returns to tertiary education rose faster than those to lower levels of schooling in urban India during the 1990s and early 2000s (Azam, 2012). As elsewhere, colleges in India have historically clustered in cities and towns. An expansion of college supply that follows this pattern risks bypassing rural areas that are less integrated with cities. The concern is especially acute in India, where rural-urban migration remains strikingly low despite large wage gaps (Munshi and Rosenzweig, 2016), making it difficult for rural youth to move to where colleges are located.

The question of who benefits has grown more urgent as higher education has expanded. Over the past half-century, higher education systems have grown worldwide, with the fastest growth in recent decades occurring in developing economies (Barro and Lee, 2013). India exemplifies this transformation. The number of colleges nearly tripled, from about 15,000 in 2003 to nearly 43,000 in 2020, and the National Education Policy 2020 aims to raise the gross enrollment ratio in higher education from 30 percent in 2023-24 to 50 percent by 2035. Tertiary attainment has risen in tandem, including in rural India: among the rural population aged 25 and older, the share with a tertiary education increased from 1.7 percent in 1991 to 7.8 percent in 2023-24, while among rural adults aged 25-34, it rose from 2.7 percent to 16.6 percent.[1] Given that colleges concentrate near towns and that rural youth rarely migrate, this sharp rise in rural attainment is somewhat puzzling. One possibility is that an expansion oriented toward towns had an unintended consequence: by placing colleges at the urban fringe, it brought them within reach of the surrounding villages.

Whether proximity alone can deliver such gains, however, is not obvious. At the primary and secondary levels, bringing educational institutions closer to where people live is one of the best-established ways to raise attainment. Building schools and reducing the distance students must travel increase enrollment and completion, as demonstrated by Indonesia's school construction program (Duflo, 2001), village-based schools in Afghanistan (Burde and Linden, 2013), and programs that reduced the effective distance to secondary schools or expanded primary schooling in India (Muralidharan and Prakash, 2017; Khanna, 2023). A growing body of evidence suggests that the same logic extends to higher education. Distance

[1] Figure 1 presents these estimates using data from the decennial Population Census and the Periodic Labour Force Survey (PLFS).

to college shapes who attends in high-income countries (Card, 1993; Currie and Moretti, 2003; Frenette, 2006; Spiess and Wrohlich, 2010), while in India, the establishment of new colleges affects local schooling markets (Jagnani and Khanna, 2020), and improved access to colleges under a central government grant scheme raised women's educational attainment, with the largest gains among rural women (Sekhri et al., 2024). Yet the decision to attend college also depends on prior preparation, expected labor market returns, and forgone earnings, none of which a nearby college can change. A new college may therefore democratize access by attracting students for whom distance had been prohibitive, or it may divert students who would have enrolled elsewhere anyway, changing where degrees are earned without changing how many are earned (Rouse, 1995; Mountjoy, 2022). Whether a large-scale, largely private expansion of conventional colleges, driven by entrants locating near towns rather than by planners allocating institutions according to need, delivers gains for rural youth remains an open question.

This paper asks whether such an expansion raised tertiary attainment in the rural villages within its reach. I study Rajasthan, one of India's largest states, which experienced a rise in rural tertiary attainment similar to that of India as a whole, alongside the fastest college expansion among Indian states. Among rural 25-34-year-olds in Rajasthan, the tertiary share rose from 2.1 percent in 1991 to 19.9 percent in 2023-24. The expansion followed regulatory changes in the early 2000s that opened higher education to private, self-financed institutions. Fewer than 60 colleges had opened in the state in any two-year period before 2003; more than 1,300 opened in the six years that followed, more than doubling the existing stock of 552. Between 2003 and 2020, the number of colleges grew at a compound annual rate of 11.3 percent, nearly double the national rate of 6.2 percent.[2] The expansion was driven overwhelmingly by private entrants. Entry was conditioned on physical inputs, above all a sizable, contiguous parcel of land, rather than on demonstrated need or quality, and new colleges clustered around existing towns. I therefore also ask what determined where the new colleges were located, and, in particular, whether predetermined structure of local landholdings determined which rural communities gained access.

Answering these questions empirically is challenging. First, the co-movement of college supply and attainment does not establish a causal link because new colleges may simply have opened where demand for higher education was already rising, as is likely near growing towns. Second, even when supply expands independently of demand, its effect on attainment is ambiguous: diversion of existing students would leave attainment unchanged, while new institutions may raise enrollment without increasing degree completion (Oppedisano, 2011). Third, little is known about how access shapes attainment at the fine spatial scale at which travel costs actually bind, particularly for women and poorer households, because existing evidence for India measures access at the district level (Sekhri et al., 2024). Finally, if colleges follow towns and land markets rather than local need, the benefits may reach villages near the urban fringe while bypassing the remote hinterland. Rajasthan offers an unusually sharp setting in which to address these questions. Because the expansion was triggered by regulatory change and carried out by private entrants, colleges reached different villages in different years within a short window. Consequently, cohorts born only a few years apart in neighboring villages faced sharply different levels of access to nearby colleges.

[2] Figure 2 and Table A1 present the corresponding numbers using data from AISHE.

Empirically identifying the effect of college expansion requires combining a precise, time-varying measure of local college access with attainment data at a comparable spatial and demographic level. I construct the former using the 2021-22 round of the All India Survey on Higher Education (AISHE), using reported establishment years to reconstruct the stock of colleges in each year and geocoding every institution to identify its precise location. Combining college locations with village centroids, I measure the number of colleges within specified distance bands of each village in Rajasthan for each year from 1998 to 2007. For attainment, I use unique cohort-level records from the Socio-Economic and Caste Census (SECC) 2011 at the village level for rural Rajasthan. Because the SECC is a census rather than a sample survey, it covers every village and allows educational attainment to be measured without sampling error. Its reporting of age in single years also permits the construction of one-year birth cohorts. I date each cohort by the year in which its members reached the typical college-entry age of 18 and assign each village-cohort observation the college access prevailing in that village in the corresponding entry year. Restricting the sample to individuals aged 22 to 31 in 2011 yields ten birth cohorts that reached college-entry age between 1998 and 2007: five before the post-2003 expansion and five during it. The result is a village-by-cohort panel that links changes in local college access to subsequent differences in tertiary attainment.

This panel allows me to compare changes in tertiary attainment across successive cohorts in villages that gained a nearby college with contemporaneous changes in villages that had not yet gained one. Because colleges reached different villages in different years, exposure is staggered. Because the effect of access plausibly builds as successive cohorts reach college-going age under improved supply, treatment effects are likely to vary with exposure time. Under these conditions, the conventional two-way fixed-effects estimator is biased (De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021). I therefore use the heterogeneity-robust staggered difference-in-differences (DiD) estimator of Callaway and Sant'Anna (2021), hereafter CS-DiD, implemented with the doubly robust estimator of Sant'Anna and Zhao (2020) and using not-yet-exposed villages as the comparison group. The main treatment is an indicator for whether at least one college lies within 15 kilometers (km) of a village when the cohort reached college-entry age.[3] The pre-expansion cohorts play a particularly useful role in this design. Their college-going decisions were made before nearby colleges opened, so their attainment could not have responded to the expansion. The pre-exposure coefficients therefore allow a direct test of whether villages that gained colleges were already on different attainment trajectories. I complement the binary design with the continuous-treatment estimator of Callaway et al. (2024), which exploits the number of colleges that entered each village's catchment.

The first main finding is that expanded local access raised tertiary attainment among exposed rural cohorts. The pre-exposure coefficients are small, individually indistinguishable from zero, and show no systematic trend, suggesting that villages that gained a nearby college and those that had not yet done so followed parallel attainment trajectories before exposure. After exposure, the effect emerges gradually. It is small and statistically insignificant for the cohort reaching college-entry age in the year access arrived. It becomes significant for

[3]I use 15 km as the baseline bandwidth because it is the distance across which the expansion moved the median village. The median distance to the nearest college fell from about 20.5 km in 1998 to about 12.3 km in 2007, while the share of villages with a college within 15 km rose from 33 percent to 61 percent. Robustness estimates using bandwidths of 10 to 14 km are similar and are reported in Subsection 5.5.

the next cohort, at about 0.7 percentage points, and rises to about 1.4-1.5 percentage points for cohorts reaching college-entry age two to three years after access arrived, and to about 2.2-2.4 percentage points four to five years after exposure. This gradual onset is consistent with attainment responding as successive cohorts reach college-going age under improved local supply. Relative to a rural tertiary attainment rate of about 6 percent among young adults in Rajasthan in 2011, these effects are economically large.

The result is robust across several dimensions. Estimates using only never-exposed villages as the comparison group are nearly identical to the baseline. A natural concern is that colleges opened near villages that were already on different development trajectories, a concern sharpened by the tendency of colleges to locate near growing towns. To address this concern, I condition on local economic activity, measured by nighttime luminosity. Because a new college can itself raise luminosity, I follow Caetano et al. (2026). I find that this concern is warranted: college access increased village luminosity by about 2.6% on average. I first condition only on pretreatment luminosity and then treat luminosity as a "bad control" whose untreated evolution is modeled explicitly. These adjustments reduce the estimated effect five periods after exposure to about 1.8 percentage points when conditioning on pretreatment luminosity and to about 1.5 percentage points in the more demanding bad-control specification, suggesting that part of the baseline estimate reflects differential local development. The effect nevertheless remains positive and statistically significant and follows the same gradual onset. Because the SECC records attainment by village of residence in 2011, migration after college-entry age could alter the measured outcome, particularly for the oldest cohorts, who had the most time to leave, and the youngest cohorts, who may not yet have completed or returned from tertiary education. Restricting the sample to cohorts reaching college-entry age between 2000 and 2005 yields estimates close to the baseline at the same horizons, rising to about 1.5 percentage points three periods after exposure. The effect also remains stable across proximity thresholds from 10 to 14 km. Finally, the continuous-treatment design delivers similar dynamics and larger magnitudes at higher doses: about 3.8 percentage points five periods after exposure for villages whose catchment gained the most colleges.

The gains in tertiary attainment accrue to both women and men. Rural tertiary attainment in Rajasthan is markedly lower for women, and distance to an institution is often thought to weigh more heavily on their schooling decisions, so I estimate the design separately by gender. The average effect is almost identical across groups: 1.25 percentage points for men and 1.24 percentage points for women. The paths differ, however. The male effect appears first, reaching about 1.5 percentage points two to three periods after exposure and 2.3-2.4 percentage points by the end of the window. The female effect builds more steadily, from about 0.5 percentage points in the first period after exposure to 2.4-3.0 percentage points four to five periods later, matching or exceeding the male effect, although the confidence bands overlap throughout. The finding that local access raises attainment for both genders indicates that proximity relaxes a constraint that binds for both groups and that the overall effect is not driven by a single group.

The second part of the paper asks where the expansion was concentrated. College entry in Rajasthan was governed by input-based requirements, of which the most binding was land: a sponsoring body needed a sizeable, contiguous parcel before it could establish a college. Because the expansion was largely private and land-intensive, the cost and availability of suitable land plausibly shaped where entrants located. I therefore relate the growth in the

number of colleges near each village between 1999 and 2007 to district-level land fractionalization in 1999. This measure is one minus the Herfindahl-Hirschman Index (HHI) of household landholdings, constructed from the National Sample Survey (NSS) 55th Round. Because it is fixed before the expansion, it cannot mechanically respond to the placement of colleges. I find that land fractionalization is strongly associated with subsequent college growth. The coefficient is positive and significant at every bandwidth from 10 to 15 km, conditional on caste composition, population density, terrain, soil quality, cropland share, night-time lights, non-farm employment, and the baseline stock of colleges. At the 10 km bandwidth, a 0.1 increase in the fractionalization index is associated with roughly 1.3 additional colleges appearing near the average village over the period. A year-by-year decomposition shows that the association emerged with the expansion itself. In 1999, the relationship between fractionalization and the number of colleges within 15 km is small and statistically indistinguishable from zero, and it remains so through 2001. It grows steadily from 2002 onward and is roughly six times its 1999 value by 2007. More- and less-fractionalized districts thus looked alike in their college endowments at the start of the period and pulled apart as the expansion unfolded. This pattern is consistent with a supply-of-land channel. Fragmented holdings are smaller, harder to mechanize, and less productive, so the underlying land carries a lower agricultural opportunity cost and offers cheaper sites for land-intensive entry. I interpret these estimates as conditional correlations rather than causal effects.

This paper contributes to three strands of the literature. The first focuses on how the supply of educational institutions shapes educational attainment. A large body of work shows that school construction and reduced distance to school raise enrollment and completion at the primary and secondary levels. Examples include the landmark evidence from Indonesia's school construction program (Duflo, 2001), village-based schools in Afghanistan (Burde and Linden, 2013), and bicycles that reduced the effective distance to secondary school for girls in India (Muralidharan and Prakash, 2017). Khanna (2023) shows, in a general-equilibrium framework, that a large-scale expansion of primary schooling in India raised educational attainment. At the tertiary level, distance to college has long been used as a source of variation in schooling (Card, 1993), and college openings in the United States have been shown to raise maternal education (Currie and Moretti, 2003). In the Indian context, much of the literature on higher education has focused on the quality margin and elite institutions. Sekhri (2020) estimates the returns to attending elite public colleges, while Jagnani and Khanna (2020) shows that the establishment of elite public colleges affects primary and secondary schooling markets in surrounding areas. Closest to this paper, Sekhri et al. (2024) studies the access margin directly. Exploiting district-level variation in college construction under a central government grant scheme, they find that greater access to colleges in home districts improved women's educational and broader development outcomes, with educational gains for women twice as large as those for men and concentrated among rural women. This paper contributes by studying a different kind of expansion: a large-scale, largely private expansion of conventional degree colleges driven by a liberalized, input-based entry regime rather than by a targeted public grant. I also estimate its effects at a much finer spatial scale, measuring access at the village level and attainment using complete-enumeration data. The census coverage of the SECC, combined with precise, time-varying measures of college proximity, allows me to trace the effects cohort by cohort and test for pre-existing differences using cohorts whose college-entry decisions predate the expansion. It also allows me to show

that the gains extend to women under a privately driven expansion, complementing the evidence of Sekhri et al. (2024) from publicly funded colleges.

The second contribution is to the literature on how urban growth affects the surrounding rural economy. A body of work shows that the growth of towns and smaller urban centers, rather than large cities alone, matters for rural welfare through labor markets, consumption linkages, and migration (Christiaensen and Todo, 2014; Gibson et al., 2017). I highlight a channel that has received little attention: the supply of higher education. Colleges established by private entrants to serve towns brought tertiary education within commuting distance of surrounding villages, raising rural attainment without requiring rural youth to migrate. In this setting, the rural gains from an urban-oriented expansion arose not by design but as a by-product of where entrants chose to locate.

The third contribution is to the literature on landholding structure and economic development. The distribution of land has been studied extensively as a determinant of agricultural productivity and the allocation of resources across sectors. Farm-size distributions and land misallocation account for a large part of cross-country differences in agricultural productivity (Adamopoulos and Restuccia, 2014; Chen et al., 2023; Chari et al., 2021), while small and fragmented holdings constrain mechanization and efficient farm sizes (Foster and Rosenzweig, 2022). Much less is known about how landholding structure shapes human capital. The existing literature on this question emphasizes land concentration as an obstacle to education. Concentrated landownership delayed the emergence of human-capital-promoting institutions (Galor et al., 2009) and slowed the expansion of schooling in Prussia (Cinnirella and Hornung, 2016), while India's landlord-based colonial tenure systems left a legacy of lower investment in public goods, including schools (Banerjee and Iyer, 2005). These studies operate primarily through the political economy of public provision and the demand for education. I highlight a different channel: land as a direct input into the supply of educational institutions. This channel connects to the literature on United States land-grant colleges, where grants of public land under the Morrill Act financed the establishment of new institutions of higher education, with lasting consequences for local innovation and agricultural productivity (Kantor and Whalley, 2019; Andrews, 2023). Whereas land-grant colleges were founded on land endowed by the state, the Rajasthan expansion was built on land acquired by private entrants. My findings suggest that, in such a setting, the predetermined structure of local landholding helps determine where new institutions are established and, hence, which communities gain access to them.

The remainder of the paper is organized as follows. Section 2 presents the expansion of higher education institutions in India, with a particular focus on Rajasthan. Section 3 details the construction of the data used in the analysis. Section 4 presents the effects of expanded college access on tertiary attainment. Section 5 reports a series of robustness checks. Section 6 presents heterogeneity in these effects by gender. Section 7 investigates land fractionalization as a potential determinant of college locations and examines whether it shaped the expansion. Finally, Section 8 concludes with a discussion of the findings.[4]

[4]In addition to the roadmap above, four appendix sections complement the main analysis. Appendix A examines whether the post-2003 expansion was specific to Rajasthan using an event study comparing the state's border districts with adjacent districts in neighboring states. Appendix B uses a random forest classifier to identify village characteristics that predict proximity to a college. Appendix C examines whether college access increased local nighttime luminosity. Appendix D presents additional tables and figures,

2 Contextual Background

The early 2000s marked a turning point for higher education in India. After two decades of steady but modest growth, the establishment of new colleges accelerated sharply. As Figure 2 shows, 2,321 colleges were established nationwide in 2003 and 2004 alone, compared with 920 in the corresponding two years a decade earlier, and the number of new establishments peaked at 4,924 in 2007-08. The national stock of colleges rose from about 15,000 in 2003 to nearly 43,000 by 2020. This acceleration took place in a changing legal and policy environment. In 2002, the Supreme Court's judgment in *T.M.A. Pai Foundation v. State of Karnataka* affirmed the right of private unaided institutions to establish and administer educational institutions, with substantial autonomy over admissions and fees, while *Islamic Academy of Education v. State of Karnataka* (2003) reiterated their freedom to set their own fee structures (Supreme Court of India, 2002, 2003). In 2003, the University Grants Commission notified regulations governing the establishment of private universities under state legislation (University Grants Commission, 2003). At the same time, national planning documents emphasized that India's low participation in higher education, along with wide rural-urban and social-group gaps, called for a large expansion of institutional capacity. Because public resources alone could not deliver this expansion, the documents assigned an increasing role to private provision (Planning Commission, Government of India, 2008). The result was an expansion driven to a large extent by self-financed private institutions.

Rajasthan entered this period with limited and geographically concentrated access to tertiary education. Fewer than 60 new colleges had opened in the state in any two-year period over the preceding two decades. Existing institutions were clustered in major cities and district headquarters (HQ), predominantly in the more developed eastern part of the state, leaving much of the large rural hinterland far from the nearest college (Department of College Education, Government of Rajasthan, 2006). With limited public resources for higher education, the state government turned to private provision. It encouraged the establishment of private, unaided colleges and self-financing courses, with the stated aim of eventually establishing a college in every tehsil (Department of College Education, Government of Rajasthan, 2006). Institutional capacity to absorb new colleges expanded in parallel. Two new affiliating universities were established in Bikaner and Kota in 2002, and by 2005-06, they had together affiliated more than 160 colleges (Department of College Education, Government of Rajasthan, 2006). The legislative framework for private education was also revised repeatedly during these years. The Rajasthan Non-Government Educational Institutions (Amendment) Ordinance was promulgated in 2003 and the Rajasthan Self-Financed Private Universities Ordinance in 2004, followed by the Rajasthan Private Universities Act of 2005 (Rajasthan Legislative Assembly, 2026).

Entry into higher education in Rajasthan was governed by input requirements. A sponsoring society or trust seeking to open a college had to secure a no-objection certificate from the state government and affiliation with a state university, and had to hold land and buildings that met the prescribed norms. The same approach was codified in detail for private universities during the study period. Under the Rajasthan Self-Financed Private Universities Ordinance of 2004 and the Rajasthan Private Universities Act of 2005, any society, public

including the robustness results discussed in Section 5.

trust, or Section 25 company could act as a sponsoring body (Government of Rajasthan, 2004, 2005). After receiving a letter of intent, the sponsor had one year to acquire a minimum of 30 acres of land and construct at least 10,000 square meters of covered space. Within the same year, it had to purchase books and journals worth at least 1 million Indian rupees (INR 10 lakh), with an undertaking to invest 5 million (INR 50 lakh) in library facilities within three years and to acquire equipment and furniture worth at least 2 million (INR 20 lakh), rising to 10 million (INR 1 crore) within five years. It also had to deposit an endowment fund with the state as security. The required endowment was 20 million (INR 2 crore) in general, 10 million (INR 1 crore) for universities located at least 15 km from the municipal limits of a divisional headquarters city or 10 km from other cities, and 7.5 million (INR 75 lakh) for such universities located in scheduled tribal areas (Government of Rajasthan, 2005). The land requirement was modest relative to those in other large states; Tamil Nadu, for instance, required at least 100 acres (Anand, 2019). Approval was thus conditioned on physical and financial inputs rather than on demonstrated educational quality. Among these inputs, land played a distinctive role. An endowment can be deposited and equipment purchased anywhere, but land is a fixed, location-specific input that must be acquired before an institution can be established, so its cost and availability plausibly influence where entry occurs.

Against this backdrop, Rajasthan experienced an expansion that was both earlier and substantially faster than the national trend. According to AISHE, the number of new colleges established in the state rose from 56 in 2001-02 to 221 in 2003-04, 399 in 2005-06, and 713 in 2007-08. Between 2003 and 2008, more than 1,300 colleges were established, more than twice the stock of 552 colleges that existed in 2003. Thus, within just a few years, the state added more colleges than the entire stock it had at the beginning of the period. Between 2003 and 2020, the stock of colleges in Rajasthan grew at a compound annual rate of 11.3 percent, nearly twice the national rate of 6.2 percent and the highest among Indian states, as shown in Appendix Table A1. This expansion was driven overwhelmingly by private entrants. Consistent with this, the comparison of Rajasthan's border districts with adjacent districts in neighboring states, presented in Appendix Figure A2, shows that the trajectories of total and privately managed colleges are nearly identical.

The expansion also changed the geography of access. New colleges appeared well beyond the established urban centers, and many villages that had no college nearby in the early 2000s gained one within a few years, as shown in Figure 3. Two features of this setting shape the analysis that follows. First, because colleges reached different villages in different years, cohorts reaching college-going age between 2003 and 2007 faced sharply different local supply depending on where they lived and when they turned eighteen. This variation underlies the estimates of the effect of access on attainment. Second, because the expansion was largely private and conditioned on inputs in which land was central, the structure of local landholding before the expansion becomes a natural candidate for explaining where new colleges were established. The next section describes the data sources and the construction of the dataset used in the analysis.

3 Data Construction

The analysis draws on two main data sources. The first is the AISHE, an annual census of higher education institutions in India conducted by the Ministry of Education, Government of India. AISHE covers universities, colleges, and standalone institutions and records their establishment year, type (e.g., affiliated college, constituent/university college, postgraduate or off-campus center, and recognized center), management (government, private-aided, or private-unaided), and location (state, district, and whether the institution is in a rural or urban area). I use the 2021-22 round. The second is a unique dataset for Rajasthan sourced from the SECC 2011, an individual-level enumeration conducted by the Government of India, with the rural enumeration led by the Ministry of Rural Development. The SECC was designed to record the socio-economic conditions and caste of households to identify deprived households for the targeting of welfare programs. For each enumerated individual, it records age, sex, and the highest level of education completed.[5] I use these two sources to measure college access and educational attainment, respectively. I supplement them with district-level data on landholdings from the 1999 NSS 55th Round and village-level data from the 2001 Population Census and the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG) (Asher et al., 2021). From SHRUG, I also use Defense Meteorological Satellite Program (DMSP) night-time lights, which are used in the robustness checks in Subsection 5.2 as a subdistrict-level covariate and in Appendix C as a village-level outcome.

For each institution, AISHE reports the year of establishment, institutional type and management, and administrative identifiers, including state, district, and institution name. I restrict the analysis to colleges, excluding universities and standalone institutions.[6] Using the reported year of establishment, I reconstruct the number of colleges in existence in each year by counting each college from its year of establishment onward. Aggregating these counts by district yields a district-year panel of the cumulative stock of colleges through 2020. I construct the same panel separately for privately managed colleges. Because district boundaries were reorganized over this period, I harmonize district identifiers to the 2001 Census boundaries by mapping districts created after 2001 back to their parent districts. This ensures that the stock of colleges in each district is measured over a consistent geographic footprint.

A district-level panel captures the expansion of college supply over time, but it is too coarse to show how access changed for individual villages. Within the same district, some villages may have gained a college within easy reach, while others remained far from one. To measure access at a finer spatial scale, I geocode every college using the Google Maps Application Programming Interface (API), querying the college's name, district, and state to recover its latitude and longitude.[7] I then match these locations to village centroids from

[5] Village-level SECC 2011 data are publicly available in PDF form. The analysis, however, uses more granular cohort-village-level data for Rajasthan.

[6] Following AISHE's classification, I focus on colleges because they deliver most undergraduate education, whereas universities differ in scale and academic scope, and standalone institutions cannot award degrees and primarily offer diploma-level technical, nursing, and teacher-training programs.

[7] Geocoding by name occasionally returns incorrect coordinates for a small fraction of colleges. I drop colleges whose latitude or longitude lies more than three standard deviations above or below the mean latitude or

the SHRUG village shapefiles. For every village in Rajasthan and each year, I count the number of colleges within radii ranging from 10 to 15 km. Using these village-year counts, I construct two measures of access: an indicator for having at least one college within a given radius and the number of colleges within that radius.

The SECC is central to this paper because, unlike a sample survey, it is a census of households. Every household in every village is enumerated, allowing village-level measures of educational attainment to be constructed without sampling error and ensuring that all villages in rural Rajasthan are observed. This level of spatial and demographic detail is not available in nationally representative household surveys. Different rounds of the NSS, PLFS, and India Human Development Survey (IHDS) sample only a limited number of households within selected villages, leaving most villages unobserved. Moreover, their publicly available data are generally not identified at the village level. The Population Census, while covering the entire population, reports educational attainment by age primarily in aggregated tables at higher administrative levels rather than in publicly available cohort-level data.

The cohort-village-level SECC data used in this paper cover Rajasthan only. Within the state, they capture the two dimensions of granularity required by the research design. First, age is recorded in single years, allowing birth cohorts to be defined at one-year intervals. Second, SECC village identifiers can be matched to the 2011 Population Census using fuzzy matching in Stata based on district and village names. This match allows cohort-level educational attainment to be aggregated to the birth-cohort and village levels and linked to the college-access measures constructed at the same fine spatial scale at which access varies.

From the SECC individual records, I classify an individual as having a tertiary education if they report graduate-level education or above. For each village and birth cohort, I compute the share of individuals with tertiary education, both overall and separately for men and women. Because educational attainment is observed only in the 2011 cross-section, I date each cohort by the calendar year in which its members reached the typical college-entry age of 18, calculated as $2011 - \text{age in } 2011 + 18$. I restrict the sample to individuals aged 22 to 31 in 2011, born between 1980 and 1989, who reached age 18 between 1998 and 2007. The five older cohorts (born 1980-1984) reached college-entry age before the post-2003 expansion and serve as the pre-exposure periods. The five younger cohorts (born 1985-1989) reached college-entry age between 2003 and 2007, during the expansion. The lower age bound ensures that even the youngest cohort had sufficient time to complete an undergraduate degree by the census date. Each village-cohort observation is then assigned the college-access measures prevailing in that village in the cohort's college-entry year, as described in Appendix Figure A3.

Next, to examine where the expansion was concentrated, I construct district-level land fractionalization in 1999 using the NSS 55th Round Employment and Unemployment Survey (1999-2000) and household-level total landholdings. Within each district of Rajasthan, I measure the concentration of rural landholdings using the HHI, defined as the sum of squared household shares of land, and define land fractionalization as one minus this index. A district in which land is concentrated among a few large holders takes a value close to zero, whereas a district in which many households each hold a small parcel takes a value approaching one. Because the measure is fixed at its 1999-2000 level, before the post-2003 expansion, it cannot mechanically respond to the placement of new colleges. I harmonize the NSS district

longitude.

identifiers to the 2001 Census boundaries and merge the measure onto villages by district, so that every village inherits the fractionalization measure of its district.[8] The dependent variable is the change in the number of colleges within each bandwidth of a village between 1999 and 2007, based on the village-year counts described above, with the 1999 count serving as the baseline control. The remaining controls are held at their pre-expansion baseline values. The Scheduled Caste and Scheduled Tribe population shares, population density, and the non-farm employment share come from the 2001 Population Census. Night-time light intensity in 2001 and the cropland share come from SHRUG. Elevation and terrain ruggedness are computed in ArcGIS. The sandy-soil share comes from the Soils of India database of the National Bureau of Soil Survey and Land Use Planning (NBSS & LUP), Indian Council of Agricultural Research (ICAR). After merging all sources, the resulting village sample comprises 42,836 villages.

Figure 2 and Figure 3 describe the expansion of colleges using AISHE data. Figure 2 plots the number of colleges established in each two-year bin from 1981 to 2020, together with the cumulative stock. Nationally, the number of establishments rose steadily through the 1990s and accelerated sharply from the early 2000s. In 2003 and 2004 alone, 2,321 new colleges were established, compared with 920 in the corresponding two-year period a decade earlier, and establishment peaked at 4,924 in 2007-08. By 2020, the cumulative stock had reached nearly 43,000 colleges. The expansion was far more abrupt in Rajasthan. Fewer than 60 colleges were established in any two-year bin before 2003. The number then jumped to 221 in 2003-04, 399 in 2005-06, and 713 in 2007-08, so that more than 1,300 colleges opened within six years, compared with a stock of 552 in 2003. Establishment slowed after 2008 but remained well above its pre-2003 level, and the stock reached 3,393 by 2020. This corresponds to a compound annual growth rate of 11.3 percent between 2003 and 2020, nearly double the national average of 6.2 percent and the highest among Indian states, as shown in Appendix Table A1. Figure 3 maps the location of every college in 2000 and 2019. In both India and Rajasthan, the expansion widened geographic coverage rather than merely densifying already-served areas: regions that were thinly served in 2000 were substantially filled in by 2019. The spread was uneven, however. In Rajasthan, the heaviest concentration by 2019 was in the eastern districts and thinned westward, with the arid western belt remaining comparatively underserved.

Figure 4 presents the share of individuals with tertiary education in Rajasthan by birth cohort, separately for the full sample, males, and females, using data from the SECC. The sample consists of individuals born between 1980 and 1989, corresponding to ages 22 to 31 in 2011. The overall share with tertiary education ranges from 4.1 percent for the 1980 cohort to 11.3 percent for the 1989 cohort. Across all cohorts, the share is higher among males than among females: among males, it ranges from 6.0 percent for the 1980 cohort to 13.9 percent for the 1989 cohort, while among females, it ranges from 1.9 percent for the 1980 cohort to 8.2 percent for the 1989 cohort. The gender gap in tertiary attainment varies between approximately 4 and 8 percentage points across cohorts, while the ratio of the male-to-female share declines from about 3.2 for the 1980 cohort to about 1.7 for the 1989 cohort. Next, I

[8] I use household-level landholding data, which are the best available for the period before 2003. However, households in these data can be identified only at the district level, as no finer geographic identifiers are available.

describe the empirical strategy, which is based on a staggered difference-in-differences design, and present the results.

4 College Access and Tertiary Educational Attainment

A larger local supply of colleges need not necessarily translate into higher local educational attainment. New institutions may attract students who would otherwise not have pursued tertiary education, but they may also enroll students who would have attended colleges elsewhere in their absence. Distinguishing between these two possibilities is therefore important in its own right. This section examines whether local birth cohorts that reached college-going age after nearby colleges were established attained higher levels of tertiary education than comparable cohorts that were not exposed to the expansion. Two features of the setting complicate a simple before-and-after comparison. First, colleges reached different villages in different years, resulting in staggered rather than simultaneous exposure. Second, the conventional two-way fixed-effects estimator can be biased when treatment timing varies, and treatment effects evolve with exposure duration, as already-treated units may serve as inappropriate controls for later-treated units (Goodman-Bacon, 2021; De Chaisemartin and d'Haultfoeuille, 2020; Sun and Abraham, 2021). I therefore estimate the effect of gaining local college access using the CS-DiD estimator of Callaway and Sant'Anna (2021), which accommodates treatment-effect heterogeneity across adoption cohorts and exposure horizons. I assess this effect in two ways: through a binary-exposure event study, in which cohorts that reached college-going age before the expansion serve as the pre-exposure periods, and through a continuous-treatment (dose-response) specification that exploits variation in exposure intensity (Callaway et al., 2024).

The analysis constructs a village-by-cohort panel from the SECC 2011, a comprehensive enumeration that records individuals' highest level of education, age, gender, and location across Rajasthan. For each village and birth cohort, I calculate the share of individuals who completed tertiary education, defined as graduate-level education or higher. This share serves as the outcome throughout the analysis.

Because educational attainment is observed only in the 2011 cross-section, I construct cohorts based on birth year and use age to identify the year in which each cohort reached the typical college-entry age. I restrict the analysis to individuals aged 22 to 31 in 2011. These individuals were born between 1980 and 1989 and reached age 18 between 1998 and 2007, respectively. Thus, the ten cohorts correspond to the following birth and college-entry years: individuals born in 1980 reached age 18 in 1998, those born in 1981 in 1999, and so on, up to those born in 1989, who reached age 18 in 2007. The five older cohorts (born in 1980-1984) reached college-entry age between 1998 and 2002, before the post-2003 expansion, and therefore serve as the pre-exposure cohorts. The five younger cohorts (born in 1985-1989) reached college-entry age between 2003 and 2007, during the expansion. By 2011, individuals in this age range were also sufficiently old for their tertiary educational attainment to have largely been completed.

This cohort construction, summarised in Appendix Figure A3, allows each village-cohort observation to be linked, through the SECC location identifiers, to the local college supply prevailing when the cohort reached the typical age of college entry. The main treatment is

an indicator for whether at least one college was located within 15 km of the village when the cohort reached college-entry age. I additionally examine alternative measures of local college access as robustness checks, including indicators for the presence of at least one college within narrower distance bands and the number of colleges located within the 10 to 15 km band. These measures distinguish between the extensive margin of college access—whether any college was present—and the intensity of local college supply. The resulting village-by-cohort panel, therefore, uses birth cohorts as the time dimension and relates variation in local college access to differences in tertiary educational attainment across cohorts exposed to the expansion.

I define exposure at the village level. Fixing a bandwidth b, let $N_{vt}(b)$ be the number of colleges within b km of village v by year t, and let

$$G_v \;=\; \min\{\, t : N_{vt}(b) \geq 1 \,\}$$

be the first cohort-year in which a college is located within b km of village v, with $G_v = 0$ for villages never reached within the sample window. Villages that already had a college within b km by the 1998 cohort are treated throughout the sample window and are excluded because they lack a pre-exposure period. The Callaway and Sant'Anna (2021) estimator compares the change in the tertiary-attainment share of villages first exposed in year g with the contemporaneous change among villages not yet exposed by year t, yielding the group-time average treatment effects on the treated:

$$ATT(g,t) \;=\; \mathbb{E}[\,Y_t - Y_{g-1} \,|\, G_v = g\,] \;-\; \mathbb{E}[\,Y_t - Y_{g-1} \,|\, G_v > t\,], \tag{1}$$

These effects are identified under a parallel-trends assumption and estimated using the doubly robust estimator of Sant'Anna and Zhao (2020), which combines an inverse-probability-weighted model for exposure with an outcome regression. Here, g indexes the adoption cohort (the year access arrived), t denotes the cohort-year in which the outcome is measured, and $g-1$ is the last pre-exposure cohort-year, which serves as the baseline for each group. I aggregate the $ATT(g,t)$ into an event study in exposure time $e = t - g$:

$$\theta(e) \;=\; \sum_{g} \Pr(G_v = g \,|\, G_v = g + e)\; ATT(g,\, g+e), \tag{2}$$

which averages the treatment effects across adoption cohorts e periods after (or before) access arrived, weighting each cohort by its share of the units observed at that exposure horizon. Post-exposure coefficients are measured relative to the last pre-exposure cohort, $g-1$, whereas each pre-exposure coefficient compares consecutive cohorts. I report 95% uniform confidence bands obtained from a multiplier bootstrap over the window $e \in [-5, 5]$. Together with the not-yet-exposed comparison group, the pre-exposure coefficients $\theta(e)$ for $e < 0$ allow the identifying assumption to be assessed directly: if villages that gained access to a nearby college followed trajectories parallel to those of villages that had not yet gained access, these coefficients should be small and show no systematic trend. Because the pre-exposure periods correspond to cohorts that reached college-entry age before nearby colleges opened, their completed tertiary education could not have responded to the subsequent expansion. These coefficients, therefore, also help assess whether the estimates reflect pre-

existing differences across villages correlated with college locations rather than the effects of the expansion itself.

4.1 Main Result

Figure 5 reports the event-study estimates for the 15 km bandwidth. The figure plots the coefficients $\theta(e)$ from Equation 2 against exposure time e, with shaded 95% uniform confidence bands. The pre-exposure coefficients ($e = -5, \ldots, -1$) are small, ranging from approximately 0.1 to 0.4 percentage points. They are individually statistically indistinguishable from zero and display no systematic trend, suggesting that villages that gained access to a nearby college and those that had not yet done so followed similar attainment trajectories before exposure. Following exposure, the effect emerges gradually. It is small and statistically insignificant in the entry year ($e = 0$), becomes statistically significant in the first period after exposure (≈ 0.7 percentage points), and increases across successive cohorts to approximately 1.4-1.5 percentage points by the second and third periods and 2.2-2.4 percentage points by the fourth and fifth periods. Each post-exposure coefficient measures the difference in the share of the cohort completing tertiary education relative to the last pre-exposure cohort and the contemporaneous change among villages not yet exposed. Thus, the coefficient of approximately 1.4 at $e = 2$ indicates that cohorts reaching college-entry age two years after their village gained access to a nearby college were about 1.4 percentage points more likely to have completed tertiary education. Relative to the low mean tertiary-attainment share at the village level, this represents an economically meaningful increase. The gradual emergence of the effect is also consistent with attainment responding over time as successive cohorts reach college-going age under improved local access. The next section presents a series of robustness checks to assess the stability of the main findings.

5 Robustness

5.1 Never-Treated Comparison Group

The main results use villages not yet exposed by period t as the comparison group. Because these villages are exposed later in the sample window, anticipation or selection in the timing of college openings could affect the comparison. Figure 6 re-estimates the same design using only villages that remain unexposed to a college within 15 km throughout the sample window as the comparison group. The pre-exposure coefficients remain small and individually insignificant, while the post-exposure effects follow the same pattern as the main results: the effect is insignificant in the entry year, becomes significant in the first period after entry, and rises to about 1.5-1.6 percentage points in the second and third periods and 2.3-2.5 percentage points in the fourth and fifth periods. The near-identical results under both comparison groups indicate that the findings are not driven by the choice of comparison group.

5.2 Pre-Treatment Covariate Adjustment

The main specification includes no covariates. A natural concern is that colleges may have opened near villages that were already on different development trajectories, violating the

unconditional parallel trends assumption. I therefore re-estimate the baseline design under a conditional parallel trends assumption, using subdistrict-level mean nighttime light intensity from the DMSP as a proxy for local economic activity. However, nightlights are not a conventional control: a new college may itself increase local luminosity through construction, electrification, and commercial activity around the campus. Appendix C confirms this: village luminosity rises by about 0.7 percent in the year of entry and by about 5 percent three periods after exposure. Conditioning on contemporaneous nightlights could therefore absorb part of the treatment effect. I instead condition only on pre-treatment nightlights. As shown by Caetano et al. (2026), this leads naturally to the estimator proposed by Callaway and Sant'Anna (2021), with pre-treatment values included as covariates. Each post-exposure $ATT(g,t)$ thus conditions on nightlights in year $g-1$, before a college opened nearby, allowing treated and not-yet-treated villages to be compared at the same pre-treatment level of local economic activity. Figure A4 shows that the pre-exposure coefficients remain small and statistically indistinguishable from zero. The post-exposure effects follow the baseline timing pattern: the effect is insignificant in the entry year and significant from the first period onward. However, the estimates are somewhat smaller, at about 1.1 percentage points in the second and third periods and 1.5-1.8 percentage points in the fourth and fifth periods.

5.3 Bad-Control Adjustment

Conditioning on pre-treatment nightlights keeps the control free of treatment effects but ignores how nightlights would have evolved between $g-1$ and t in the absence of a college. If this untreated evolution differs across villages in ways that also affect tertiary attainment, parallel trends conditional on pre-treatment nightlights alone may fail. Appendix C shows that luminosity responds to changes in college access and therefore cannot be treated as a predetermined covariate. The second approach in Caetano et al. (2026) addresses this concern by conditioning on the untreated potential value of nightlights in period t, that is, the level a village would have reached had no college opened nearby. Because this value is unobserved for treated villages, I identify it under a covariate unconfoundedness assumption: conditional on pre-treatment nightlights and the lagged outcome, the untreated evolution of nightlights is independent of treatment timing. This approach generalizes the one in the previous subsection but requires an additional model for the evolution of nightlights. I estimate $ATT(g,t)$ using not-yet-treated villages as the comparison group. Figure A5 shows that the pre-exposure coefficients remain small and statistically indistinguishable from zero. The post-exposure effect is insignificant in the entry year and the first period, becomes statistically significant from the second period onward at about 0.9 percentage points, and rises to about 1.5 percentage points by the fifth period. The effect is therefore smaller than the baseline estimate, suggesting that part of the baseline effect may reflect differential local development. However, it remains positive and statistically significant, with a similar gradual onset.

5.4 Restricted Cohort Window

Because the SECC records educational attainment by village of residence in 2011, migration between the year a cohort reached college-entry age and the census date could affect the

measured outcome. This concern is greatest at the two ends of the cohort window. The oldest cohorts, who reached college-entry age in 1998 and 1999, had the most time to leave their villages after completing their education, for example, for employment or through marriage. The youngest cohorts, who reached college-entry age in 2006 and 2007, were closest to the census date and may not yet have completed tertiary education or returned to their villages after studying elsewhere. In both cases, the share of tertiary-educated individuals observed in a village could reflect who remained there rather than who completed tertiary education. To limit this concern, I re-estimate the baseline design using a narrower window that retains only cohorts reaching college-entry age between 2000 and 2005, dropping the two oldest and two youngest cohorts. The estimator, the 15 km proximity measure, and the not-yet-exposed comparison group remain unchanged. Villages already exposed by the time the 2000 cohort reached college-entry age are excluded because they have no pre-exposure period. The event study is reported over the window $e \in [-3, 3]$.

Figure A6 reports the results. The post-exposure pattern closely matches the baseline. The effect is small and statistically insignificant in the entry year (about 0.4 percentage points). It becomes statistically significant in the first period after exposure (about 0.9 percentage points) and rises to about 1.2 percentage points in the second period and 1.5 percentage points in the third, close to the baseline estimates at the same horizons. The pre-exposure coefficients at $e = -2$ and $e = -1$ are small and statistically indistinguishable from zero. The coefficient at $e = -3$ is positive (about 0.7 percentage points), and its 95% uniform confidence band just includes zero. Two features limit the relevance of this estimate for assessing the identifying assumption. First, in the restricted window, the coefficient at $e = -3$ is identified only from villages first exposed in 2004 and 2005 and compares the two earliest cohorts retained in the sample. It therefore reflects a single change at the edge of the event window rather than the experience of treated villages as a whole. Second, it does not form part of a trend leading up to exposure: the coefficients at $e = -2$ and $e = -1$ are close to zero and have opposite signs. Because this coefficient raises the average of the pre-exposure estimates in this specification, I interpret the restricted-window results as evidence on the role of migration rather than as an additional test of parallel trends. The latter is assessed using the full cohort window in Figure 5. The fact that the post-exposure effect persists after removing the cohorts most exposed to post-schooling migration suggests that the baseline estimates are not driven by changes in the residential composition of villages between college-entry age and the census date.

5.5 Alternative College Distance Thresholds

To confirm that the choice of 15 km is innocuous, I re-estimate the main specification using alternative bandwidths. Figure A7 presents the event-study results for bandwidths ranging from 10 to 14 km, with each panel replicating the analysis in Figure 5 using a different proximity threshold. The qualitative pattern remains stable across bandwidths: small pre-exposure coefficients are followed by a positive post-exposure effect that is statistically significant from the first period and grows with exposure time, reaching approximately 2.1-2.4 percentage points by the fourth period across all bandwidths. At bandwidths of 10 to 12 km, the fifth-period coefficient is somewhat smaller, at approximately 1.5-1.8 percentage points, but remains positive and statistically significant. Because the longest exposure horizon is iden-

tified only from the earliest-exposed villages, these estimates rely on fewer adoption cohorts and are therefore less precisely estimated. The pre-exposure coefficients are individually insignificant at bandwidths of 12 to 14 km. At bandwidths of 10 and 11 km, the coefficient at $e = -2$ is positive and marginally significant, but the remaining pre-exposure coefficients are close to zero and show no systematic trend. Overall, the results suggest that the estimated effect is not an artifact of a particular proximity threshold.

5.6 Continuous-Treatment DiD

The binary indicator classifies a village as exposed once a single college enters its catchment, thereby ignoring the extent to which physical access expands. This distinction matters because the number of colleges within a village's catchment increased substantially over the study period. I therefore complement the binary design with the continuous-treatment DiD estimator of Callaway et al. (2024). The dose is defined as the number of colleges within b km of the village, $N_v(b)$, rescaled by its cross-village maximum:

$$D_v \;=\; \frac{N_v(b)}{\max_{v'} N_{v'}(b)} \;\in\; [0,1]. \tag{3}$$

Under this normalization, $D_v = 0$ for a village with no college in its catchment, while $D_v = 1$ for the village with the largest number of colleges within its catchment. The group-time parameters are then defined conditional on the dose:

$$ATT(g,t \mid d) \;=\; \mathbb{E}[\,Y_t - Y_{g-1} \,|\, D_v = d,\; G_v = g\,] \;-\; \mathbb{E}[\,Y_t - Y_{g-1} \,|\, G_v > t\,]\,. \tag{4}$$

As in the binary specification, I aggregate these estimates into an event study in exposure time, using not-yet-treated villages as the comparison group, and report the dose-response evaluated at the maximal dose. Figure 7 presents the estimates for the 15 km bandwidth. The pattern resembles that of the binary specification. The pre-exposure coefficients are small, ranging from approximately 0.2 to 0.4 percentage points, and none is statistically significant. The coefficient in the period of exposure is approximately 0.5 percentage points and is only marginally distinguishable from zero. The estimates then rise steadily. The effect becomes statistically significant in the first period after exposure, at approximately 1.0 percentage point. It increases to approximately 1.8 percentage points in the second period, 2.2 in the third, 3.2 in the fourth, and 3.8 in the fifth, all evaluated at the maximal dose. The larger magnitudes relative to the binary estimates reflect the difference between the two exposure measures. The continuous specification evaluates the dose-response at the highest dose, whereas the binary specification averages across villages with different levels of exposure.

Figure A8 shows that the dose-response pattern is broadly similar across bandwidths ranging from 10 to 14 km. In all five specifications, the post-exposure estimates become statistically significant from the first period, at approximately 0.9-1.0 percentage points. They reach approximately 1.7-1.9 percentage points by the second period and 2.8-3.3 percentage points by the fourth period. In the fifth period, the estimates diverge somewhat. At 13 and 14 km, they continue to rise, reaching approximately 3.5-3.7 percentage points. At 10-12

km, they fall back to approximately 2.4-2.7 percentage points, with wider confidence intervals. At these narrower bandwidths, the coefficient at $e = -2$ is positive, at approximately 0.7-1.0 percentage points, and statistically significant, while the remaining pre-exposure coefficients are small and generally insignificant. Despite this isolated pre-exposure coefficient, the post-exposure estimates indicate a substantially larger and sustained increase. By the fourth period, their magnitudes are roughly three times that of the coefficient at $e = -2$.

Taken together, the robustness exercises point to a consistent and economically meaningful relationship between college expansion and tertiary educational attainment. Both the binary and continuous-treatment specifications indicate that cohorts reaching college-entry age after local access expansion attained tertiary education at higher rates, with effects that increased with time since exposure. Under the baseline 15 km specification, the binary estimates rise to approximately 1.4 percentage points two periods after exposure and 2.2-2.4 percentage points after four to five periods. The continuous-treatment estimates indicate larger gains at the maximal dose, reaching approximately 3.2 percentage points by the fourth period and 3.8 percentage points by the fifth. This pattern is consistent with larger gains in tertiary attainment in villages with greater exposure to college expansion.

These findings are robust across a range of alternative specifications, including the use of a never-treated comparison group, restrictions on the cohort window to address concerns about migration, and alternative distance thresholds. The pre-exposure estimates are small and, with few exceptions, statistically insignificant, whereas the post-exposure estimates show a clear and sustained increase. Adjusting for pre-treatment local economic activity or accounting for the untreated evolution of night-time luminosity moderates the estimated magnitudes but leaves a positive and statistically significant effect after exposure. Overall, the results consistently show that the expansion of local college access increased tertiary attainment among cohorts reaching college-entry age after the expansion. The next section examines heterogeneity by gender, investigating whether and to what extent the effects of college expansion on tertiary attainment differ between men and women.

6 Heterogeneity by Gender

The main results capture the overall attainment response, combining men and women when calculating village-level tertiary attainment. However, the two groups need not respond to improvements in local college access in the same way. Rural tertiary attainment in Rajasthan is substantially lower among women than among men in these cohorts. Moreover, distance to an institution may impose a greater constraint on women's educational decisions. A local expansion of physical access could therefore narrow, widen, or leave unchanged the gender gap in tertiary attainment.

To examine this heterogeneity, I re-estimate the binary-exposure design specified in Equation 1 and Equation 2 separately for men and women. The estimator, the 15 km proximity measure, the cohort construction, and the not-yet-exposed comparison group remain unchanged. The only difference is the outcome variable, Y_{vt}, which is recalculated separately by gender as the share of men or women in each village-cohort cell who have completed tertiary education.

Figure 8 reports the results for men. The pre-exposure coefficients ($e = -5, \ldots, -1$)

are small, individually indistinguishable from zero, and show no discernible trend. This pattern is consistent with parallel attainment trajectories before access to a nearby college among male cohorts that gained access and those that were not yet exposed. Following exposure, the estimated effect is close to zero in the entry year ($e = 0$), becomes statistically significant in the first post-exposure period (approximately 0.8 percentage points), reaches about 1.5 percentage points in the second and third periods, and rises to approximately 2.3-2.4 percentage points by the fourth and fifth periods.

Figure 9 presents the corresponding results for women. The pre-exposure coefficients are again small and show no discernible trend, consistent with parallel pre-exposure attainment trajectories. The post-exposure effect is statistically insignificant in the entry year but becomes significant in the first post-exposure period (approximately 0.5 percentage points). It then increases steadily, reaching about 1.2 percentage points in the second period, 1.6 percentage points in the third, and 2.4-3.0 percentage points by the fourth and fifth periods.

Two features stand out. First, for both men and women, the attainment response is confined to cohorts of college-going age when access improved and is absent among pre-expansion cohorts, whose schooling was already complete. This is precisely the pattern expected under a causal interpretation and one that college placement alone would not generate. Second, both genders respond from the first period after exposure. The male effect is somewhat larger in the early periods, while the female effect rises more steadily and reaches a similar or larger magnitude by the fourth and fifth periods, although the confidence bands overlap throughout. The finding that improved access to local colleges raises measured tertiary attainment for women as well as men, rather than benefiting only the group already more likely to enrol, is consistent with proximity relaxing a constraint that binds for both genders. It also indicates that the pooled effect reported in Section 4 is not driven by a single gender. Having established that local access raised tertiary attainment, I next ask where the new colleges were established, and in particular whether the pre-expansion distribution of land, measured by district land fractionalization in 1999, shaped their placement.

7 Land Fractionalization and the Expansion of Colleges

Where were the new colleges established, and why were they established there? The expansion need not have been geographically even—some districts absorbed far more of the new capacity than others—and understanding the patterns of college placement helps establish whether the rollout was shaped by pre-existing local conditions rather than educational need.

Section 2 motivates examining land first. Entry in Rajasthan was governed by a relatively permissive but explicitly input-based regime: a sponsoring body could secure approval largely by meeting physical and financial thresholds, the most binding of which was land. The framework required a minimum of 30 contiguous acres, along with a covered construction requirement of 10,000 square meters. Acquiring a sizable parcel of suitable land was therefore a precondition for entry rather than an incidental cost. Because the expansion was overwhelmingly private, self-financed, and land-intensive, the cost and availability of land plausibly shaped where entrants located, largely independently of local educational demand or student preparedness. This regulatory feature motivates my measure: if land is a first-order, quasi-fixed input to college entry, a predetermined index of the local structure of

landholding should help predict where the expansion took hold.

I use district land fractionalization in 1999, defined as one minus the HHI of household land shares; higher values indicate a greater prevalence of smallholders and more fragmented landholdings. Fragmented holdings tend to be smaller, harder to mechanize, and less productive, so the underlying land may have a lower agricultural opportunity cost and a lower price. Faced with a binding acreage requirement, cost-sensitive entrants had reason to seek out precisely such low-value land. Districts with more fragmented holdings may therefore have offered cheaper sites for land-intensive expansion. Two features make the measure attractive as a potential determinant. First, it is fixed at its 1999-2000 level, before the post-2003 expansion, so it cannot mechanically respond to college placement. Second, as I show, it predicts subsequent college growth relative to a balanced 1999 baseline rather than merely capturing the pre-existing stock.

I relate the geography of the expansion to baseline land structure using a village-level first-difference specification. Because the predetermined covariates—land fractionalization and the control vector—are time-invariant (fixed at their 1999 or 2001 baseline), a standard fixed-effects or first-difference transformation would eliminate them from the specification. To allow these covariates to shape changes in the local college stock, I begin with a two-period reduced-form specification in which these covariates are interacted with time. Let $D_t = \mathbb{1}[t = 2007]$ denote the endpoint of the expansion window, and write, for the number of colleges within b km of village v,

$$C^b_{v,t} \;=\; \mu_v \;+\; \lambda D_t \;+\; \beta_{1999} F_{d(v)} \;+\; \beta_\Delta \left(F_{d(v)} \times D_t\right) \;+\; \mathbf{X}'_v \delta_{1999} \;+\; \left(\mathbf{X}_v \times D_t\right)' \delta_\Delta \;+\; u_{v,t}, \quad (5)$$

where μ_v is a village fixed effect that absorbs any permanent, location-specific propensity to host colleges, including any fixed correlation between local land structure and college siting; λ is a common period effect; $F_{d(v)}$ denotes land fractionalization in district $d(v)$ in 1999; and $\mathbf{X}_v$ is a vector of predetermined controls: Scheduled Caste and Scheduled Tribe population shares, population density, elevation, terrain ruggedness, cropland share, nighttime lights, sandy-soil share, and nonfarm employment share, all measured at their 2001 baseline. The coefficients β_Δ and δ_Δ on the time interactions are the objects of interest: they capture how the relationships between the respective covariates and the local college stock change between the two endpoints.

Taking the difference between the two periods for each village eliminates the fixed effect μ_v and the period-invariant level terms $\beta_{1999} F_{d(v)}$ and $\mathbf{X}'_v \delta_{1999}$, leaving only the interaction coefficients:

$$\Delta C^b_v \;\equiv\; C^b_{v,2007} - C^b_{v,1999} \;=\; \alpha \;+\; \beta\, F_{d(v)} \;+\; \gamma\, C^b_{v,1999} \;+\; \mathbf{X}'_v \delta \;+\; \varepsilon_v, \quad (6)$$

where $\alpha \equiv \lambda$, $\beta \equiv \beta_\Delta$, and $\delta \equiv \delta_\Delta$. Equation 6 is the specification I estimate: a single cross-section of villages in which the differenced college count is regressed on 1999 land fractionalization and the time-interacted baseline controls, with standard errors clustered at the district level. I additionally condition on the baseline stock $C^b_{v,1999}$ to account for mean reversion and path dependence in college provision—the one term not generated by the differencing itself.

Differencing the dependent variable is central to my approach. Without the baseline-stock control, the coefficient β in Equation 6 would capture the land fractionalization ×

time interaction, $\beta = \beta_{2007} - \beta_{1999}$, identified from the change in the fractionalization-college gradient, net of the village fixed effect μ_v. Conditioning on $C^b_{v,1999}$ makes β the change in this gradient among villages with the same initial college stock. It therefore isolates college growth that is not explained by mean reversion or path dependence in college provision. The specification thus asks not whether villages in more fractionalized districts have more colleges, but whether they accumulated colleges at a faster rate during the expansion.

To trace the underlying gradient directly, I unpack the two endpoints into the full sequence of years, estimating the cross-sectional level relationship separately for each year at the 15 km bandwidth:

$$C^{15}_{v,t} \;=\; \alpha_t \;+\; \beta_t\, F_{d(v)} \;+\; \mathbf{X}'_v\delta_t \;+\; \varepsilon_{v,t}, \qquad t = 1999, \ldots, 2007, \tag{7}$$

so that each β_t represents the period-t slope on land fractionalization in Equation 7. The endpoint contrast, $\beta_{2007} - \beta_{1999}$, corresponds to the first-difference coefficient without the baseline-stock control. Examining the full path β_t reveals whether this divergence is a genuine post-2003 phenomenon. Under the logic outlined above, the path should be flat and statistically indistinguishable from zero early in the window, before rising thereafter.

Table 1 reports estimates of Equation 6 across bandwidths from 10 to 15 km. Land fractionalization enters positively and is statistically significant at the 5% level across all bandwidths. The point estimates rise monotonically with the radius—from 13.07 at 10 km to 25.40 at 15 km—as wider catchments encompass more of the new institutions. At the 15 km bandwidth, the estimate of 25.40 implies that a 0.1 increase in the fractionalization index is associated with roughly 2.54 additional colleges appearing within 15 km of the average village over the period. The coefficient on the baseline college count is positive, precisely estimated, and exceeds across all bandwidths, rising from 1.50 to 1.95. This pattern indicates strong path dependence: villages already well served in 1999 accumulated colleges fastest, consistent with agglomeration in the establishment of new institutions.

The preceding results establish a positive relationship between baseline land fractionalization and college expansion. Next, I examine how this relationship evolves over the expansion period by tracing the association between land fractionalization and college provision year by year. Table 2 reports estimates of Equation 7 separately for each year from 1999 to 2007. The pattern is clear. In 1999, the association is small and statistically indistinguishable from zero (7.50, standard error 4.97), and it remains statistically insignificant through 2001. From 2002 onward, the coefficient rises steadily and becomes statistically significant at the 10% level, reaching 47.53 by 2007—roughly a sixfold increase relative to its 1999 value. Fractionalized and non-fractionalized districts thus had similar college endowments at the start of the period but progressively diverged thereafter. This timing indicates that the positive estimates in Table 1 reflect a change in the gradient rather than a fixed baseline association: the relationship between land fractionalization and college provision emerges progressively over the expansion period.

The timing and regulatory backdrop point to a common determinant operating through land supply. Because entry required assembling a substantial parcel, land costs were a binding constraint on siting, and fragmented, lower-value holdings offered the inexpensive land sought by entrants pursuing a land-intensive, self-financed expansion. Read in this light, the result speaks directly to the paper's central finding: colleges opened where land

was cheaper rather than where demand and preparedness were strongest, thereby expanding physical access.[9]

Two caveats are worth noting. First, the estimates are conditional correlations. Although the balanced 1999 baseline and the monotone post-2002 divergence are reassuring, they do not, by themselves, identify a causal effect of fractionalization on college growth. Second, the supply-of-land channel is one determinant consistent with the evidence, not the only one. Agrarian conditions that increase fractionalization may also shift household demand toward nonfarm pathways, while fragmented ownership could, in principle, raise the transaction costs of assembling a contiguous parcel. The regressions establish the empirical regularity, but the underlying determinant remains a matter of interpretation. The next section concludes by discussing the implications of these findings and directions for future research.

8 Conclusion and Discussion

This paper studies one of the largest and fastest expansions of higher education in India, the post-2003 growth of colleges in Rajasthan, and asks whether expanded local access to colleges raised tertiary attainment among the cohorts exposed to it. In answering this question, it also examines where the new colleges were established. The answers rest on two complementary data sources. The first is the universe of colleges from AISHE, which I geocode to measure year-by-year access for every village in Rajasthan. The second is the cohort-level data from the SECC 2011, a complete enumeration that measures tertiary attainment in every village for each single-year birth cohort. Linking the two yields a village-by-cohort panel of ten cohorts, aged 22 to 31 in 2011, who reached college-entry age between 1998 and 2007. In this panel, cohorts that reached college-going age before the expansion serve as a natural benchmark for those that reached it afterward.

The first finding is that expanded local access raised tertiary attainment among exposed cohorts. Cohorts that reached college-entry age after a college opened within 15 km of their village were more likely to complete tertiary education than comparable cohorts in villages not yet reached, by about 1.4 percentage points two periods after access arrived and by 2.2-2.4 percentage points after four to five periods. Relative to a rural tertiary attainment rate of about 6 percent among young adults in Rajasthan in 2011, these effects are large. The dynamics of the response are as informative as its size. The effect is statistically indistinguishable from zero for the cohort reaching college-entry age in the year access arrived. This is the pattern one would expect if attainment can respond only as successive cohorts reach

[9] Appendix B presents complementary village-level evidence on placement. A random forest classifier trained on 31 geographic, demographic, land-use, and infrastructure characteristics predicts whether a village lay within 15 km of a college by 2011, with a held-out area under the receiver operating characteristic (ROC) curve (AUC) of 0.96. Distance to the nearest statutory town, latitude, longitude, nighttime lights, subdistrict urbanization, and distance to the district headquarters together account for roughly 80 percent of total permutation importance. Village literacy, caste composition, and the nonfarm employment share jointly account for about 4 percent, while local amenities contribute almost nothing, as shown in Figure A9. These patterns indicate that new colleges were more closely associated with proximity to existing towns and economic activity than with measures of local educational demand or need. Because the predictors are measured in 2011, some may partly reflect the presence of colleges themselves (see Appendix C), so I treat this exercise as descriptive.

college-going age under improved local supply. It is difficult to reconcile this pattern with a story in which colleges simply opened in villages already on rising trajectories, because the pre-exposure cohorts, whose schooling was complete before nearby colleges opened, show small coefficients and no systematic trend. The estimates are stable when only never-exposed villages are used as the comparison group, across proximity thresholds from 10 to 14 km, and when the sample is restricted to cohorts reaching college-entry age between 2000 and 2005 to limit concerns about migration. Adjusting for local economic activity, measured by nighttime luminosity, moderates the magnitudes. The estimates fall to about 1.5-1.8 percentage points four to five periods after exposure, suggesting that part of the baseline effect reflects differential local development. The effect nonetheless remains positive and statistically significant, with the same gradual onset. The continuous-treatment design adds a dose-response dimension: villages whose catchment gained more colleges saw larger gains, reaching about 3.2 percentage points after four periods and 3.8 after five periods at the maximum dose.

The effect is significant for both women and men, and the average effect is almost identical across the two groups (1.25 and 1.24 percentage points). Both groups respond from the first period after exposure, but their paths differ. The effect for men is somewhat larger initially, at about 1.5 percentage points two to three periods after exposure, while the effect for women rises more steadily, from about 0.5 percentage points in the first period to 2.4-3.0 percentage points by the fourth and fifth periods, matching or exceeding the effect for men, although the confidence bands overlap throughout. Rural women in Rajasthan began from markedly lower levels of tertiary attainment, and distance to an institution is often thought to be a stronger constraint on their schooling decisions. The finding that local supply raised attainment for both groups indicates that the cost of reaching a college was a genuine constraint for rural youth. It also suggests that the expansion did more than reallocate enrollment among students who would have completed a degree in any case, since diversion alone would have left village-level attainment unchanged.

The second finding concerns where the expansion took hold. District land fractionalization in 1999, measured before the expansion began, is positively and significantly associated with subsequent growth in the number of colleges near a village at every bandwidth from 10 to 15 km, conditional on caste composition, population density, terrain, soil quality, cropland share, nighttime lights, nonfarm employment, and the baseline stock of colleges. At the 10 km bandwidth, a 0.1 increase in the fractionalization index is associated with about 1.3 additional colleges near the average village over the period, compared with about 2.5 at 15 km. The timing of this relationship is particularly revealing. In 1999, more- and less-fractionalized districts had essentially similar college endowments, and the association remains small and statistically indistinguishable from zero through 2001. It then grows steadily from 2002 onward, reaching roughly six times its 1999 value by 2007. The gradient between land structure and college provision was thus not a preexisting feature of the landscape but emerged with the expansion itself. The baseline stock of colleges also strongly predicts subsequent growth, pointing to path dependence and agglomeration in the places where new institutions were established. The village-level prediction exercise also points in the same direction: proximity to existing towns, the district headquarters, and economic activity predicts which villages had a college nearby by 2011, whereas literacy, caste composition, and the nonfarm employment share together contribute very little. Considered

alongside an entry regime in which acquiring a sizeable parcel of land was a precondition for establishing a college, these patterns are consistent with a land-supply channel. Fragmented, smaller, and less productive holdings carry a lower agricultural opportunity cost and therefore plausibly offer cheaper sites for a land-intensive, largely private expansion.

Taken together, the two findings show that the expansion of higher education affected both the location of institutions and the educational attainment of rural youth. Proximity to a college raised the likelihood that rural youth completed a degree, while the location of new colleges was related to the structure of local landholding and proximity to existing towns rather than to measures of local educational need. The geography of human capital accumulation in this setting was therefore shaped not only by the characteristics of students and households but also by supply-side conditions, including the local land market, which influenced where institutions could be established. The two findings are also linked. Because new colleges followed towns and economic activity, the adjustments for nighttime luminosity matter, and the fact that a significant effect persists after these adjustments supports interpreting the attainment estimates as a response to access rather than to local development. This evidence extends the literature on landholding structure, which has emphasized its implications for agricultural productivity, sectoral change, and the political economy of public schooling. Here, land also enters directly into the geography of educational institutions. In this respect, the Rajasthan expansion contrasts with the land-grant colleges of the United States: whereas those institutions were founded on land endowed by the state, the colleges studied here were established by private entrants who had to secure land, making the availability of suitable land an important consideration in determining where they were established. From a policy perspective, the results suggest that an input-based, largely private entry regime can raise rural tertiary attainment, including among women, but that where it delivers access depends on where entrants find it cheap to locate. The arid western districts of Rajasthan, for example, remained comparatively underserved even after the expansion. This finding is relevant to how states pursue the National Education Policy 2020 target of a gross enrollment ratio of 50 percent by 2035, although this paper does not evaluate the costs, fees, or quality of the institutions involved.

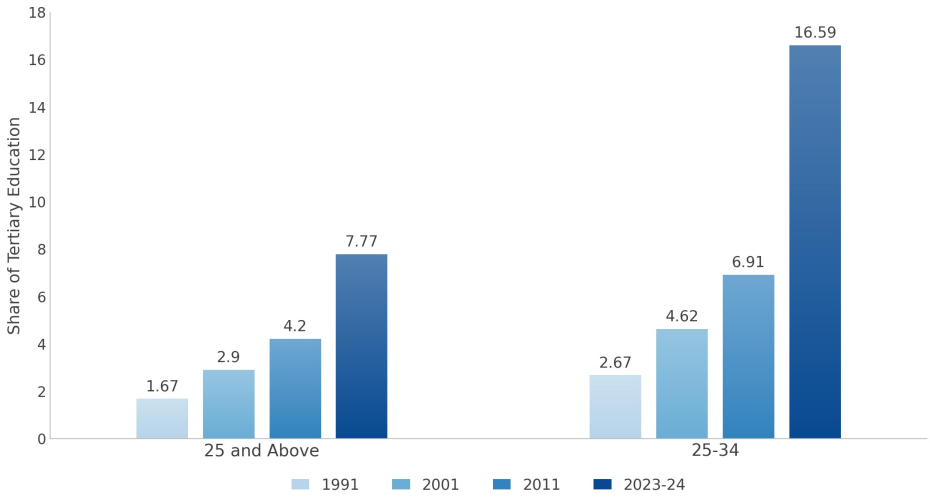
Several features of the analysis should be kept in mind when interpreting these results. The attainment estimates are reduced-form effects of local college access. They are not estimates of the effect of enrollment, and they are measured using the village of residence in 2011, so they could be affected by migration after college-entry age, although the results for the restricted cohort window suggest that migration does not drive the findings. The college stock is reconstructed from establishment years in the 2021–22 AISHE round, so institutions that opened and closed before that round are not captured. A few pre-exposure coefficients at narrower bandwidths are marginally significant, although they show no systematic trend. The land-fractionalization estimates are conditional correlations, measured at the district level because the household landholding data do not identify a finer geography. Their timing is consistent with, but does not by itself establish, a causal relationship between local landholding structure and the location of new colleges. Other factors may also be at work, including agrarian conditions that shift household demand toward nonfarm pathways, and the village-level evidence on proximity to towns is descriptive because its predictors are measured in 2011.

The analysis also points to several directions for future research. If more granular

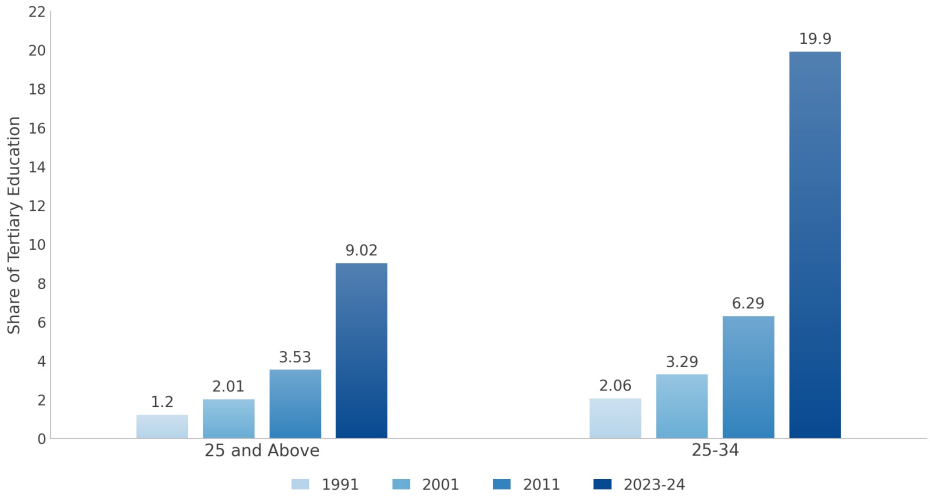
individual-level and institutional data become available, it would be possible to follow exposed cohorts into the labor market and examine whether additional degrees translated into higher earnings, nonfarm employment, or occupational mobility, and whether these returns differed between men and women. Such data could also allow researchers to examine the quality of the institutions established during the expansion and whether it varied with the conditions under which they entered. More detailed information on students' places of origin and destinations could further reveal the extent to which new colleges served local youth or attracted students from other villages and districts. Finally, comparable granular data from other states could be used to examine whether the relationship between landholding structure and the geography of higher education shown here extends beyond Rajasthan. These extensions would help clarify the longer-run effects and broader relevance of the rapid expansion of private higher education examined in this paper.

Figures and Tables

Figure 1: Rural Tertiary Education Attainment, 1991 to 2023-24



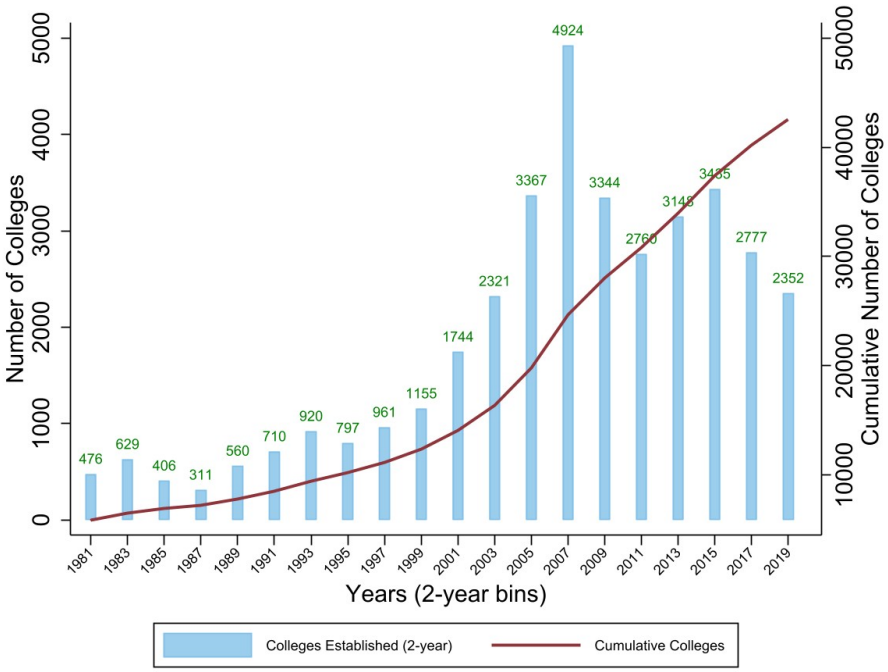
(a) India



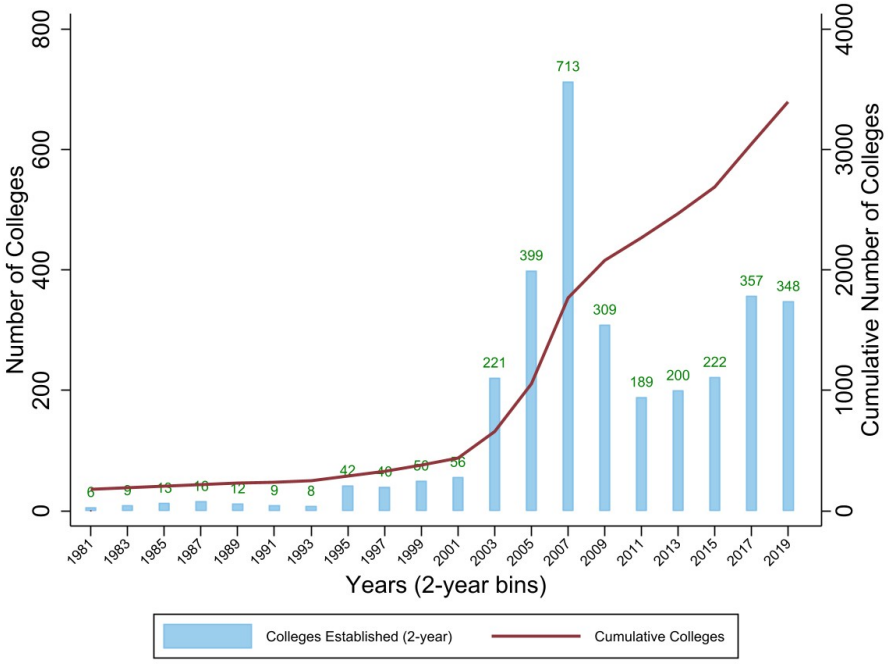
(b) Rajasthan

Notes: Bars show the rural share of the population with tertiary education for two age groups, 25 and above and 25-34, across 1991, 2001, 2011, and 2023-24. Data for 1991, 2001, and 2011 are from the Population Census, while data for 2023-24 are from the PLFS.

Figure 2: College Establishment over Time, 1981 to 2020



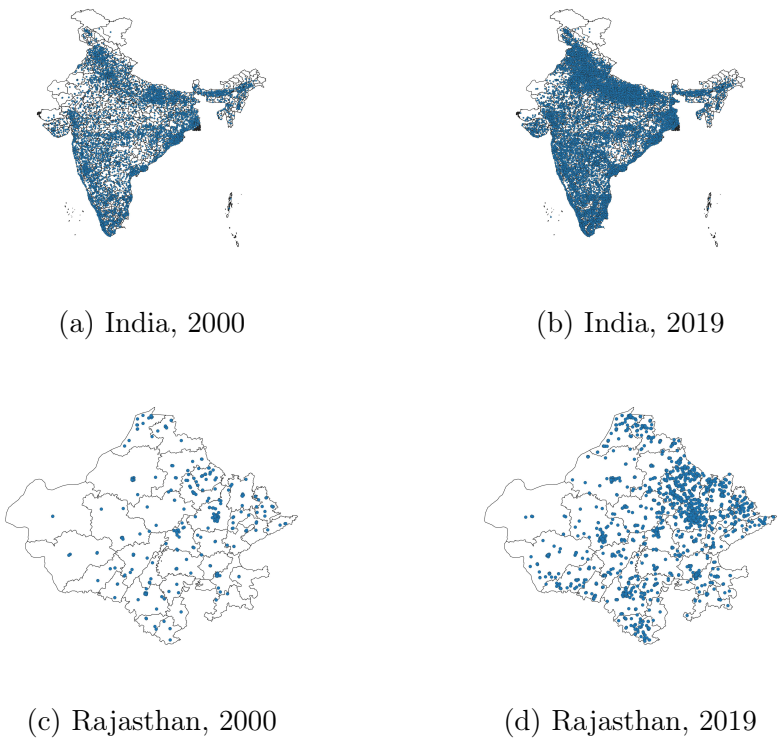
(a) India



(b) Rajasthan

Notes: Source - AISHE, Ministry of Education, Government of India, 2021-22. Bars show the number of colleges established in each two-year bin from 1981 to 2020. The line shows the cumulative stock of colleges, including institutions established before 1981.

Figure 3: Spatial Distribution and Expansion of Colleges Over Two Decades

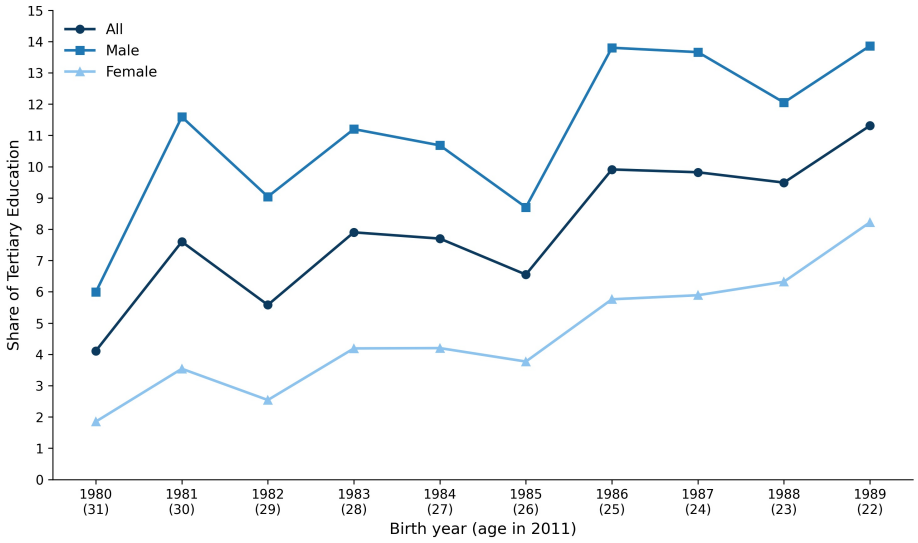


(a) India, 2000 (b) India, 2019

(c) Rajasthan, 2000 (d) Rajasthan, 2019

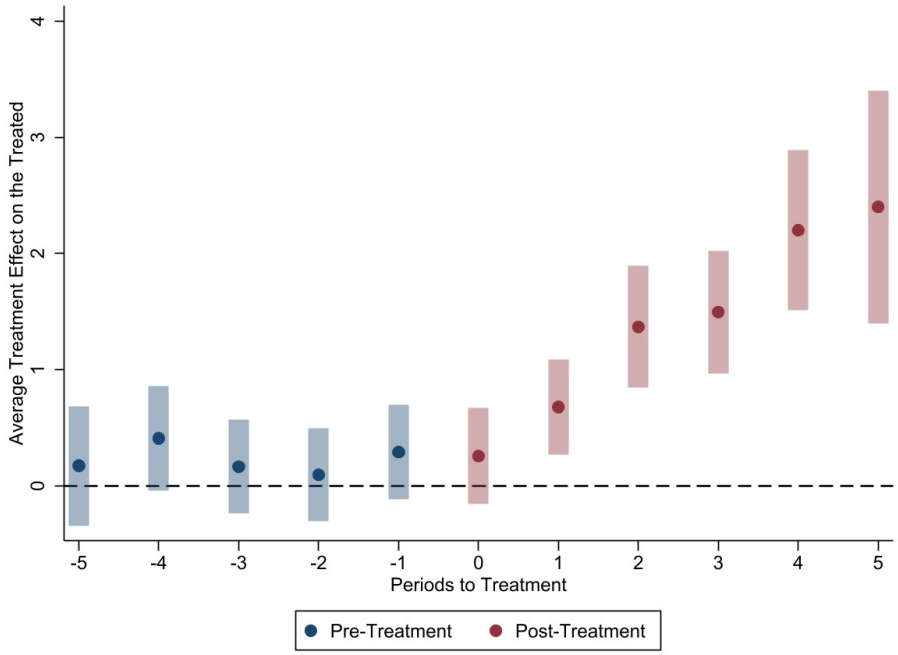
Notes: Source - AISHE, Ministry of Education, Government of India, 2021-22. Each blue dot represents a college. The top row shows the all-India distribution, while the bottom row shows Rajasthan. The left and right columns correspond to 2000 and 2019, respectively.

Figure 4: Rajasthan Share of Tertiary Education by Birth Cohort and Gender



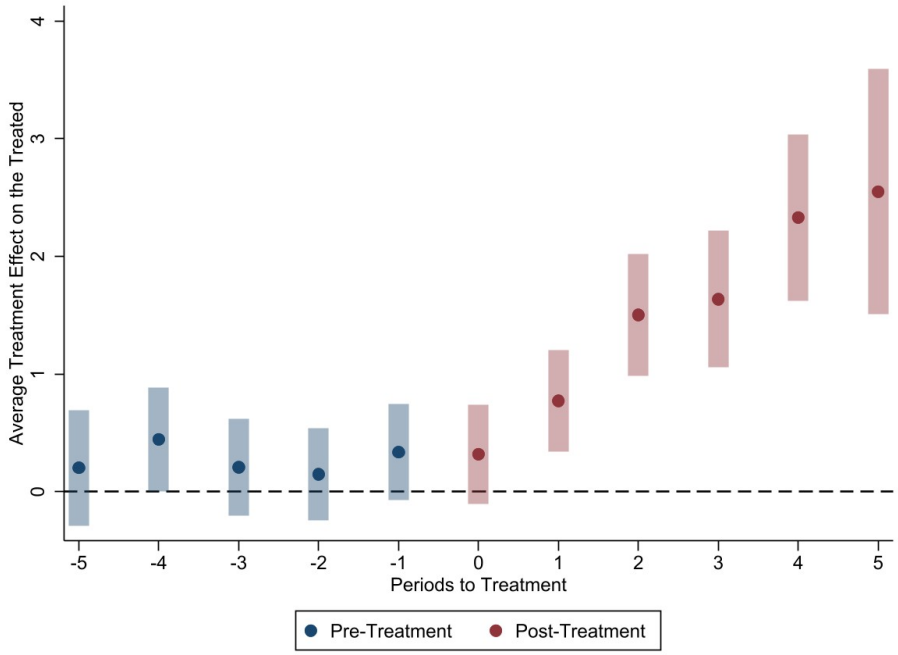
Notes: Source - SECC, 2011. The figure plots the share of individuals in Rajasthan with tertiary education by birth year, separately for the full sample, males, and females. Birth year is calculated as 2011 minus the age reported in 2011. The sample includes individuals aged 22 to 31 in 2011.

Figure 5: CS-DiD Estimates at the 15 km Bandwidth



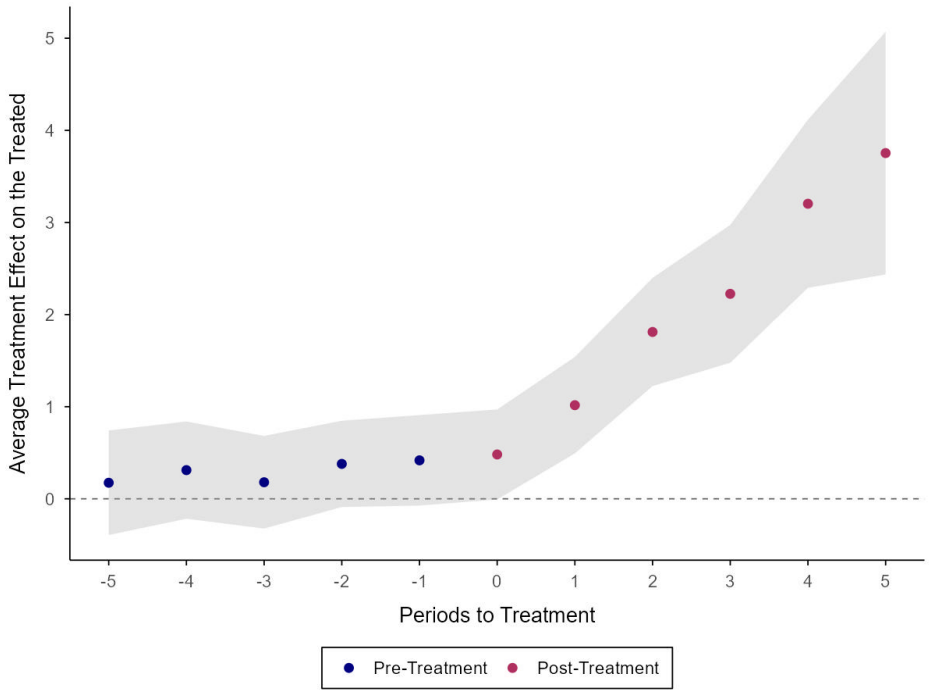
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure 6: CS-DiD Estimates at the 15 km Bandwidth, Never-Treated Comparison Group



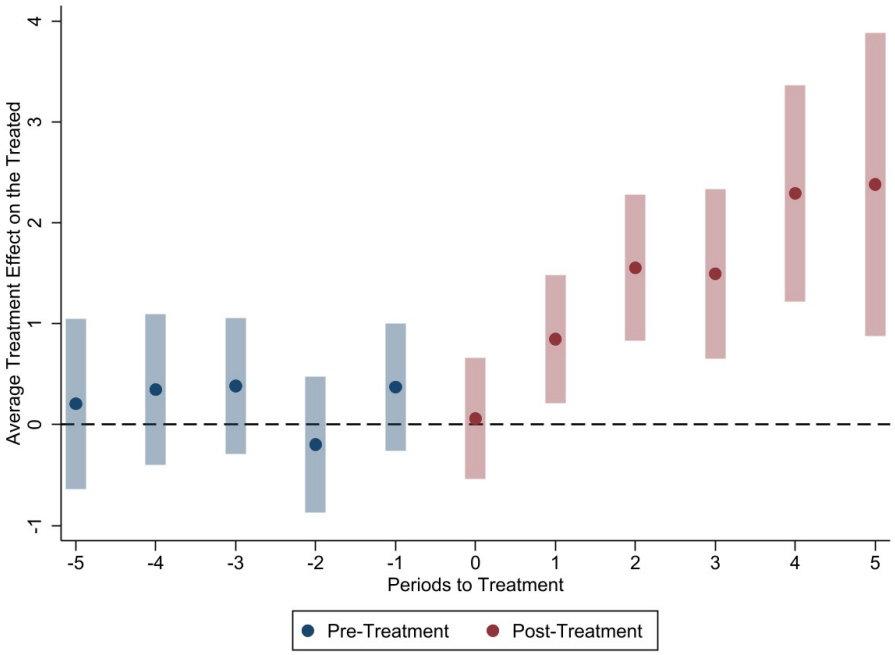
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), using never-treated villages as the comparison group. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure 7: Continuous CS-DiD Estimates at the 15 km Bandwidth



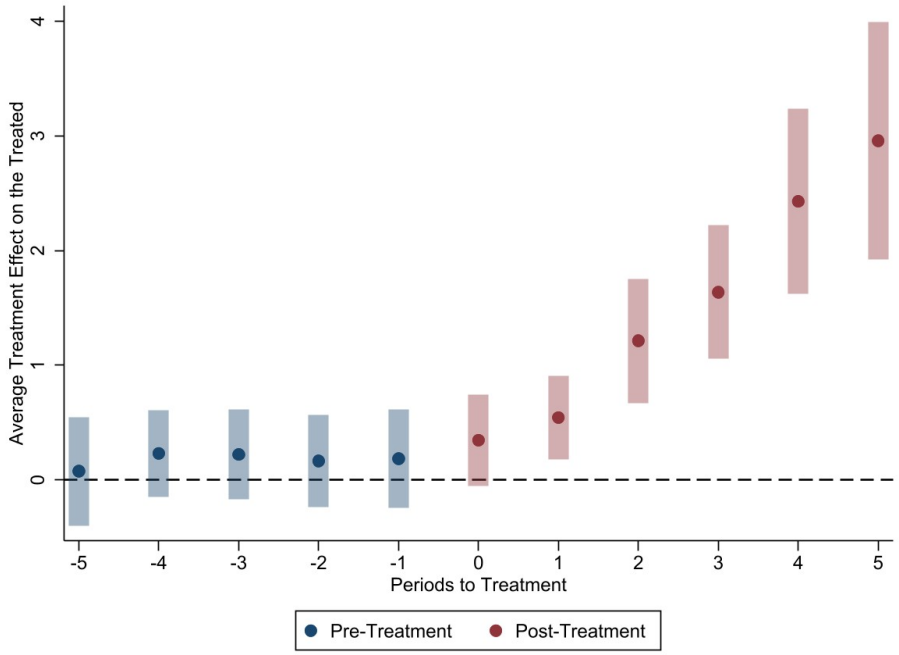
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from continuous-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator, with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure 8: CS-DiD Estimates at the 15 km Bandwidth, Males



Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of males completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure 9: CS-DiD Estimates at the 15 km Bandwidth, Females



Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of females completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Table 1: Land Fractionalization and Growth in College, 1999-2007

	Change in number of colleges, 1999-2007					
	(1)	(2)	(3)	(4)	(5)	(6)
Bandwidth	10 km	11 km	12 km	13 km	14 km	15 km
Land fractionalization (1999)	13.067**	15.406**	17.998**	20.434**	23.280**	25.396**
	(5.946)	(7.162)	(8.199)	(9.296)	(10.429)	(11.298)
Baseline colleges (1999)	1.497***	1.606***	1.693***	1.793***	1.888***	1.950***
	(0.154)	(0.173)	(0.193)	(0.221)	(0.246)	(0.241)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	42,836	42,836	42,836	42,836	42,836	42,836
R-Squared	0.471	0.502	0.532	0.568	0.604	0.640

Notes: Robust standard errors clustered at the district level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the change in the number of colleges within the stated bandwidth of a village between 1999 and 2007. Land fractionalization was measured at the district level in 1999. Baseline colleges (1999) are the number of colleges within the same bandwidth in 1999. Controls are the share of the Scheduled Caste population, the share of the Scheduled Tribe population, population density, elevation, terrain ruggedness, sandy-soil share, cropland share, night-time lights, and the share of non-farm employment. The Scheduled Caste and Scheduled Tribe shares, population density, and the non-farm employment share are sourced from the 2001 Census; night-time lights and cropland share from SHRUG; elevation and terrain ruggedness are computed in ArcGIS; and the sandy-soil share from the Soils of India database (NBSS & LUP, ICAR).

Table 2: Land Fractionalization and College Density (15 km), by Year

	Number of colleges within 15 km								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Year	1999	2000	2001	2002	2003	2004	2005	2006	2007
Land fractionalization (1999)	7.502	9.584	10.813	12.726*	17.305*	22.840*	28.648*	35.777*	47.525*
	(4.972)	(5.811)	(6.433)	(7.300)	(9.868)	(12.708)	(16.003)	(20.687)	(25.419)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	42,836	42,836	42,836	42,836	42,836	42,836	42,836	42,836	42,836
R-Squared	0.179	0.183	0.177	0.169	0.172	0.172	0.179	0.186	0.204

Notes: Robust standard errors clustered at the district level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The dependent variable is the number of colleges within 15 km of a village in the year indicated by the column; each column is a separate cross-sectional regression. Land fractionalization was measured at the district level in 1999. Controls are the share of the Scheduled Caste population, the share of the Scheduled Tribe population, population density, elevation, terrain ruggedness, sandy-soil share, cropland share, night-time lights, and the share of non-farm employment. The Scheduled Caste and Scheduled Tribe shares, population density, and the non-farm employment share are sourced from the 2001 Census; night-time lights and cropland share from SHRUG; elevation and terrain ruggedness are computed in ArcGIS; and the sandy-soil share from the Soils of India database (NBSS & LUP, ICAR).

References

Adamopoulos, T. and Restuccia, D. (2014). The size distribution of farms and international productivity differences. *American Economic Review*, 104(6):1667–1697.

Anand, A. (2019). Private universities act: Different states, different norms. Careers360, 5 April 2019. https://news.careers360.com/private-universities-act-different-states-different-norms.

Andrews, M. J. (2023). How do institutions of higher education affect local invention? evidence from the establishment of us colleges. *American Economic Journal: Economic Policy*, 15(2):1–41.

Asher, S., Lunt, T., Matsuura, R., and Novosad, P. (2021). Development research at high geographic resolution: an analysis of night-lights, firms, and poverty in india using the shrug open data platform. *The World Bank Economic Review*, 35(4):845–871.

Azam, M. (2012). Changes in wage structure in urban india, 1983–2004: A quantile regression decomposition. *World Development*, 40(6):1135–1150.

Banerjee, A. and Iyer, L. (2005). History, institutions, and economic performance: The legacy of colonial land tenure systems in india. *American economic review*, 95(4):1190–1213.

Barro, R. J. and Lee, J. W. (2013). A new data set of educational attainment in the world, 1950–2010. *Journal of development economics*, 104:184–198.

Burde, D. and Linden, L. L. (2013). Bringing education to afghan girls: A randomized controlled trial of village-based schools. *American Economic Journal: Applied Economics*, 5(3):27–40.

Caetano, C., Callaway, B., Payne, S., and Sant'Anna, H. (2026). Difference-in-differences with "bad controls". *arXiv preprint arXiv:2608.03881*.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. NBER Working Paper 32117, National Bureau of Economic Research.

Callaway, B. and Sant'Anna, P. H. (2021). Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230.

Card, D. (1993). Using geographic variation in college proximity to estimate the return to schooling. NBER Working Paper 4483, National Bureau of Economic Research.

Castelló-Climent, A., Chaudhary, L., and Mukhopadhyay, A. (2018). Higher education and prosperity: From catholic missionaries to luminosity in india. *The Economic Journal*, 128(616):3039–3075.

Castelló-Climent, A. and Mukhopadhyay, A. (2013). Mass education or a minority well educated elite in the process of growth: The case of india. *Journal of Development Economics*, 105:303–320.

Chari, A., Liu, E. M., Wang, S.-Y., and Wang, Y. (2021). Property rights, land misallocation, and agricultural efficiency in china. *The Review of Economic Studies*, 88(4):1831–1862.

Chen, C., Restuccia, D., and Santaeulàlia-Llopis, R. (2023). Land misallocation and productivity. *American Economic Journal: Macroeconomics*, 15(2):441–465.

Christiaensen, L. and Todo, Y. (2014). Poverty reduction during the rural–urban transformation–the role of the missing middle. *World Development*, 63:43–58.

Cinnirella, F. and Hornung, E. (2016). Landownership concentration and the expansion of education. *Journal of Development Economics*, 121:135–152.

Currie, J. and Moretti, E. (2003). Mother's education and the intergenerational transmission of human capital: Evidence from college openings. *The Quarterly Journal of Economics*, 118(4):1495–1532.

De Chaisemartin, C. and d'Haultfoeuille, X. (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American economic review*, 110(9):2964–2996.

Department of College Education, Government of Rajasthan (2006). Status, policies and initiatives in higher education in rajasthan. https://collegeeducation.rajasthan.gov.in/httpdocs/eprc.htm.

Duflo, E. (2001). Schooling and labor market consequences of school construction in indonesia: Evidence from an unusual policy experiment. *American economic review*, 91(4):795–813.

Foster, A. D. and Rosenzweig, M. R. (2022). Are there too many farms in the world? labor market transaction costs, machine capacities, and optimal farm size. *Journal of Political Economy*, 130(3):636–680.

Frenette, M. (2006). Too far to go on? distance to school and university participation. *Education Economics*, 14(1):31–58.

Galor, O., Moav, O., and Vollrath, D. (2009). Inequality in landownership, the emergence of human-capital promoting institutions, and the great divergence. *The Review of economic studies*, 76(1):143–179.

Gibson, J., Datt, G., Murgai, R., and Ravallion, M. (2017). For india's rural poor, growing towns matter more than growing cities. *World Development*, 98:413–429.

Glaeser, E. L. and Maré, D. C. (2001). Cities and skills. *Journal of labor economics*, 19(2):316–342.

Goodman-Bacon, A. (2021). Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277.

Government of Rajasthan (2004). The rajasthan self-financed private universities ordinance, 2004. https://indiankanoon.org/doc/119903960/.

Government of Rajasthan (2005). The rajasthan private universities act, 2005 (act no. 10 of 2005). https://indiankanoon.org/doc/150590448/.

Jagnani, M. and Khanna, G. (2020). The effects of elite public colleges on primary and secondary schooling markets in india. *Journal of Development Economics*, 146:102512.

Kantor, S. and Whalley, A. (2019). Research proximity and productivity: long-term evidence from agriculture. *Journal of Political Economy*, 127(2):819–854.

Khanna, G. (2023). Large-scale education reform in general equilibrium: Regression discontinuity evidence from india. *Journal of Political Economy*, 131(2):549–591.

Moretti, E. (2004a). Estimating the social return to higher education: evidence from longitudinal and repeated cross-sectional data. *Journal of econometrics*, 121(1-2):175–212.

Moretti, E. (2004b). Workers' education, spillovers, and productivity: evidence from plant-level production functions. *American Economic Review*, 94(3):656–690.

Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review*, 112(8):2580–2630.

Munshi, K. and Rosenzweig, M. (2016). Networks and misallocation: Insurance, migration, and the rural-urban wage gap. *American Economic Review*, 106(01):46–98.

Muralidharan, K. and Prakash, N. (2017). Cycling to school: Increasing secondary school enrollment for girls in india. *American Economic Journal: Applied Economics*, 9(3):321–350.

Oppedisano, V. (2011). The (adverse) effects of expanding higher education: Evidence from italy. *Economics of Education Review*, 30(5):997–1008.

Planning Commission, Government of India (2008). Eleventh five year plan 2007–12, volume ii: Social sector. https://www.niti.gov.in/sites/default/files/2023-08/11th_vol2.pdf.

Psacharopoulos, G. and Patrinos, H. A. (2018). Returns to investment in education: a decennial review of the global literature. *Education Economics*, 26(5):445–458.

Rajasthan Legislative Assembly (2026). The rajasthan ordinances. https://assembly.rajasthan.gov.in/Containers/Legislation/Ordinance.aspx.

Rouse, C. E. (1995). Democratization or diversion? the effect of community colleges on educational attainment. *Journal of Business & Economic Statistics*, 13(2):217–224.

Sant'Anna, P. H. and Zhao, J. (2020). Doubly robust difference-in-differences estimators. *Journal of econometrics*, 219(1):101–122.

Sekhri, S. (2020). Prestige matters: Wage premium and value addition in elite colleges. *American Economic Journal: Applied Economics*, 12(3):207–225.

Sekhri, S., Hossain, M. A., and Khosla, P. (2024). Access to colleges, human capital, and empowerment of women. *Journal of Human Resources*, 59(2):502–544.

Spiess, C. K. and Wrohlich, K. (2010). Does distance determine who attends a university in germany? *Economics of Education Review*, 29(3):470–479.

Sun, L. and Abraham, S. (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of econometrics*, 225(2):175–199.

Supreme Court of India (2002). T.m.a. pai foundation v. state of karnataka, (2002) 8 scc 481. https://indiankanoon.org/doc/512761/.

Supreme Court of India (2003). Islamic academy of education v. state of karnataka, (2003) 6 scc 697. https://indiankanoon.org/doc/1978528/.

University Grants Commission (2003). Ugc (establishment of and maintenance of standards in private universities) regulations, 2003. https://www.ugc.gov.in/oldpdf/pub/newsletters/jan_2004/16.htm.

Valero, A. and Van Reenen, J. (2019). The economic impact of universities: Evidence from across the globe. *Economics of education review*, 68:53–67.

Appendices

A College Expansion in Rajasthan

Did the supply of colleges in Rajasthan expand disproportionately after 2003, and if so, when did that acceleration begin? An increase in the number of colleges in Rajasthan need not reflect anything specific to the state—it could simply mirror a broader, nationwide expansion of higher education over the same period. To isolate a Rajasthan-specific shock, I compare districts on Rajasthan's side of the state border with otherwise similar districts just across it: contiguous districts that share broadly similar geography, agro-climatic conditions, and economic environments, but fall under different state administrations. Because these neighboring districts are exposed to the same national trends in college supply but differ in which state governs them, common national trends can be netted out, and a discrete gap that opens up between them can be interpreted as specific to Rajasthan. I therefore ask whether Rajasthan's border districts experienced a discrete increase in their local stock of colleges relative to adjacent districts across the border, and when that divergence emerged. I answer this question using an event-study specification on the district-year college panel from AISHE, restricting attention to contiguous border districts and taking 2003 as the base year.

This analysis uses the district-year panel and restricts the sample to border districts of six states—Gujarat, Haryana, Madhya Pradesh, Punjab, Rajasthan, and Uttar Pradesh—defined using 2001 Census districts to ensure consistent geographic boundaries. The analysis is restricted to 22 Rajasthan border districts and 26 districts from the five neighboring states (5 from Gujarat, 7 from Haryana, 10 from Madhya Pradesh, 2 from Punjab, and 2 from Uttar Pradesh), yielding 48 districts in total.[10] Restricting the comparison to geographically proximate districts on either side of Rajasthan's border limits confounding from broad regional differences in economic development, culture, and pre-existing educational infrastructure while preserving a balanced treated-control comparison. Figure A1 shows the border map of the treated and control groups based on the 2011 administrative boundaries available in SHRUG.

I begin by establishing that the border sample captures the college expansion described above, and that the shock occurs only in the post-2003 period. Using the district-year panel, restricted to 1995 to 2010 (16 years) and constructed from AISHE, I estimate the following event-study specification:

$$Y_{dt} = \alpha_d + \lambda_t + \sum_{k=1995,\ k\neq 2003}^{2010} \beta_k \left(\mathrm{RJ}_d \times \mathbf{1}[t = k]\right) + \varepsilon_{dt}, \tag{A1}$$

where Y_{dt} is the cumulative number of colleges established in district d up to year t; RJ_d is an indicator equal to one for Rajasthan border districts; α_d and λ_t are district and year fixed effects; and standard errors are clustered at the district level. The year 2003 is the omitted base period, so each β_k measures the differential college stock in Rajasthan border

[10] According to the 2011 Population Census, because of bifurcation, there were 49 districts, 23 of which were in Rajasthan.

districts relative to adjacent control districts in year k, net of common national trends and permanent district-level differences. The coefficients for $k \leq 2002$ allow me to assess parallel pre-trends directly: if the two sides of the border followed similar institutional trajectories before the expansion, these coefficients should be jointly indistinguishable from zero.

Figure A2 reports the estimates. In both panels, the pre-2003 coefficients are small, cluster just below zero, and display no pronounced trend, with 95% confidence intervals that include zero throughout the pre-period—indicating that Rajasthan border districts and their neighbors were accumulating colleges at statistically similar rates before the expansion. Beginning in 2004, the coefficients turn positive and rise steadily throughout the post-period. For overall colleges shown in Figure A2a, the differential reaches roughly thirty-three additional colleges per district relative to the control group by 2010, becoming statistically significant at the 5% level within one to two years of the expansion and remaining so through the end of the sample window, even as the confidence intervals widen in the final years. The private-college estimates presented in Figure A2b trace an almost identical trajectory, reaching approximately thirty-one additional colleges per district by 2010. The near-coincidence of the two paths indicates that the expansion was driven overwhelmingly by the private sector, which accounts for essentially all of the additional college stock accumulated in Rajasthan's border districts after 2003.[11]

These district-level patterns are consistent with the aggregate college growth numbers presented in Table A1: Rajasthan recorded a compound annual growth rate in the number of colleges of 11.3 percent between 2003 and 2020, nearly double the national rate of 6.2 percent and higher than that of each neighboring state in the control group. Collectively, the event studies confirm that the border sample isolates a sharp, discretely timed supply shock to tertiary access that begins precisely after 2003 and is absent in the pre-period.

B Predictors of College Proximity

The expansion of tertiary education institutions in Rajasthan during the 2000s brought hundreds of new colleges to a state marked by sharp geographic contrasts, from the densely settled eastern plains to the sparsely populated Thar Desert in the west. A natural descriptive question is where this expansion occurred—which village characteristics best predict proximity to a newly established college? To answer this question, I construct a dataset of 43,136 villages and define a binary outcome equal to one if a village lies within 15 km of a college established by 2011, and zero otherwise.[12] The predictor set comprises 31 village characteristics drawn from the Village Directory of the 2011 Census, satellite-derived mea-

[11] The gap between the overall and private trajectories—on the order of two colleges per district by 2010—reflects the limited contribution of government colleges to the post-2003 expansion.

[12] The outcome is constructed from the distance to the nearest such college, using a 15 km proximity cutoff. Roughly two-thirds of villages (about 67 percent) are classified as proximate, so the two classes are moderately imbalanced. The two samples differ because they draw on different data sources. The prediction exercise uses village characteristics from the 2011 Census Village Directory and satellite-derived measures, yielding a sample of 43,136 villages. The analysis in Section 7 combines data from the 2001 Population Census, SHRUG, the Soils of India database, and the district-level NSS land fractionalization measure, retaining 42,836 villages that could be matched across all these sources.

sures of terrain and economic activity, and geographic coordinates.[13] These variables span four domains: (i) geography and terrain, comprising latitude, longitude, elevation, surface roughness, and sandy soil cover; (ii) demographic and socioeconomic composition, comprising population density, the sex ratio, male and female literacy rates, Scheduled Caste and Scheduled Tribe population shares, and the share of non-farm employment; (iii) land use and economic activity, comprising cropland and forest cover shares, night-time light intensity, and the subdistrict urbanization rate; and (iv) local infrastructure and market access, comprising the availability of schools, all-weather roads, national and state highways, bus service, electricity, markets, commercial banks, and primary health centres, together with distances to the district headquarters and the nearest statutory town.

The relationship between village characteristics and college proximity is unlikely to be linear. To maximize predictive accuracy, I model the probability of proximity using a random forest classifier, an ensemble learning method that flexibly captures nonlinearities and high-order interactions without imposing a parametric functional form, making it particularly well suited to descriptive prediction exercises of this kind. The forest comprises 1,000 trees, with the square root of the number of predictors sampled at each split and a minimum leaf size of five villages. I evaluate predictive performance on a stratified 25 percent hold-out sample and, in addition, through stratified five-fold cross-validation. The model sharply discriminates between proximate and non-proximate villages: the held-out AUC is 0.962, essentially identical to the five-fold cross-validated AUC of 0.964 (standard deviation 0.002), while the precision-recall AUC is 0.982. The predicted probabilities are reasonably well calibrated, with a Brier score of 0.079. Predicted and observed proximity frequencies lie close to the 45-degree line in the upper part of the probability range, where most villages are concentrated, while the model somewhat overpredicts proximity in the lowest probability bins, as shown in Figure A9a. Observable village characteristics thus account for a substantial share of the variation in college proximity, even though placement retains a genuinely idiosyncratic component.

To identify which characteristics drive these predictions, I compute permutation importance, measured as the reduction in test-set AUC when each predictor is randomly shuffled. Figure A9b plots the ten leading predictors; the shares reported below are expressed relative to total permutation importance. The results reveal a striking pattern: proximity to newly established colleges is governed primarily by physical and administrative geography and by the location of existing economic activity, rather than by the socioeconomic characteristics of villages. Distance to the nearest statutory town is the single most important predictor, reducing test-set AUC by 0.063 and accounting for 26.3 percent of total permutation importance, followed by latitude (16.1 percent), longitude (12.3 percent), night-time light intensity (10.3 percent), the subdistrict urbanization rate (8.4 percent), and distance to the district headquarters (6.2 percent). Collectively, these six variables account for roughly 80 percent of total permutation importance. Sandy soil cover, elevation, cropland share, and surface

[13] As a first-best approach, I would rely exclusively on post-expansion characteristics. Because this is a descriptive prediction exercise rather than a causal one, and because most predictors are geographic, terrain-based, or otherwise slow-moving, I use contemporaneous 2011 characteristics. A small number of economic-activity measures, such as night-time lights and urbanization, may partly reflect the colleges themselves (see Appendix C) and should therefore be interpreted as associations rather than as exogenous determinants.

roughness round out the top ten, together contributing a further 16 percent. In contrast, characteristics that a demand-driven or equity-oriented placement rule would be expected to prioritize play almost no role. The seven demographic and socioeconomic composition variables—male and female literacy, Scheduled Caste and Scheduled Tribe shares, the non-farm employment share, population density, and the sex ratio—jointly account for just 4.0 percent of total importance, with each contributing less than 1 percent individually. Local infrastructure and amenities contribute essentially nothing: the availability of schools at any level, all-weather roads, national and state highways, bus service, electricity, markets, banks, and primary health centers each account for 0.1 percent of total importance or less, with several registering permutation importance indistinguishable from zero.

Collectively, these findings indicate that the spatial footprint of the expansion was shaped by where towns and economic activity already existed, rather than by the characteristics of the villages reached by the expansion. The dominance of distance to the nearest statutory town, distance to the district headquarters, and night-time light intensity indicates that new colleges were concentrated in and around existing urban and administrative centers rather than dispersed into the rural interior. The strong roles of latitude, longitude, elevation, and sandy soil cover reflect Rajasthan's pronounced settlement gradient: in the arid, sparsely populated west, villages are mechanically farther from towns and institutions, making geographic location and terrain powerful predictors of proximity. Equally informative is what does not predict proximity: village-level human capital, caste composition, and local amenities carry little explanatory power, indicating that placement was neither systematically targeted toward educationally disadvantaged or marginalized communities nor driven by stronger local demand or complementary infrastructure. Overall, the evidence suggests that the expansion largely reinforced existing spatial and administrative structures, with college locations determined by proximity to towns, economic activity, and physical geography rather than by village-level socioeconomic conditions or educational need.

C College Access and Local Night-Time Luminosity

Subsection 5.2 and Subsection 5.3 use night-time luminosity as a proxy for local economic activity and treat it as a "bad control" because a new college may itself raise luminosity through construction, electrification, and commercial activity around the campus. Appendix B also shows that luminosity is one of the leading predictors of college proximity. This appendix examines whether college access increased luminosity in villages that gained access. If so, conditioning on contemporaneous luminosity could absorb part of the treatment effect, motivating the adjustments in Subsection 5.3.

I use DMSP night-time lights from SHRUG, matched to 2011 Census villages, for 1998 to 2007. These years coincide with the college-entry years in the main panel (Appendix Figure A3). The outcome is $\ln(1 + \text{mean luminosity})$, so the estimated coefficients approximate proportional changes. The treatment definition, comparison group, and estimator follow Subsection 4.1. Treatment is an indicator for at least one college within 15 km of a village, the comparison group consists of not-yet-exposed villages, and estimation uses the doubly robust estimator of Sant'Anna and Zhao (2020). Villages already exposed in 1998 are excluded because they have no pre-exposure period. I aggregate the group-time estimates into

an event study over $e \in [-5, 5]$ using a wild bootstrap.

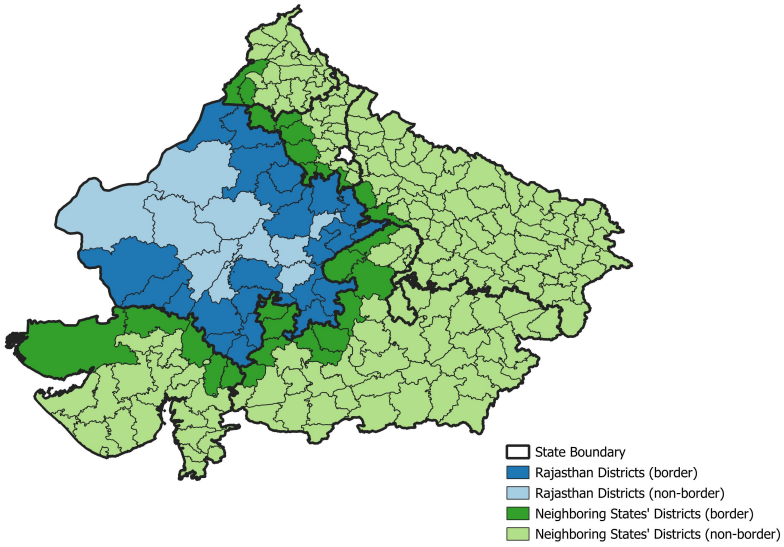
Figure A10 reports the results. Before exposure, the coefficients at $e = -4$, $e = -2$, and $e = -1$ are close to zero and statistically insignificant. In contrast, the coefficients at $e = -5$ and $e = -3$ are positive, at about 1.3 and 1.2 percent, respectively, and statistically significant. These estimates qualify the parallel-trends interpretation, although their magnitudes are small relative to the post-exposure peak. After exposure, luminosity rises by about 0.7 percent in the entry year, with marginal statistical significance. The estimated increase reaches about 2.1 percent in the first period and 3.5 percent in the second, peaks at about 5.0 percent in the third, and then declines to about 3.5 and 2.9 percent in the fourth and fifth periods, respectively.

These results inform the main analysis in two ways. First, luminosity responds to college access earlier than tertiary attainment. The luminosity effect is detectable in the entry year, whereas the attainment effect in Subsection 4.1 becomes statistically significant only in the first period after exposure. This timing is consistent with construction and local economic activity preceding the arrival of the first exposed cohort of students. The findings support treating luminosity as a potentially post-treatment variable rather than a predetermined covariate, motivating the approach in Subsection 5.3. Second, the two outcomes follow different trajectories. Luminosity peaks in the third period and subsequently declines, whereas the attainment effect continues to rise through the fifth period. This divergence suggests that the dynamics of the attainment effect are not simply mirrored by those of luminosity. However, the difference in trajectories alone cannot rule out luminosity as a channel affecting the estimated attainment effects.

Three caveats warrant attention. First, DMSP data are coarse in rural areas because of blooming and top-coding, making village-level luminosity a noisy proxy for economic activity. Second, the estimates cannot distinguish lighting at the campus itself from broader local development. Third, the statistically significant coefficients at $e = -5$ and $e = -3$ indicate departures from zero in some pre-exposure periods and therefore qualify the parallel-trends interpretation. Although these coefficients are small relative to the post-exposure peak, their statistical significance means that the pre-exposure evidence does not fully support the parallel-trends assumption. I therefore interpret these results as evidence that luminosity responds to college access, rather than as a precise estimate of the effect of colleges on local economic activity. The analysis also does not establish whether luminosity growth affects tertiary attainment or vice versa.

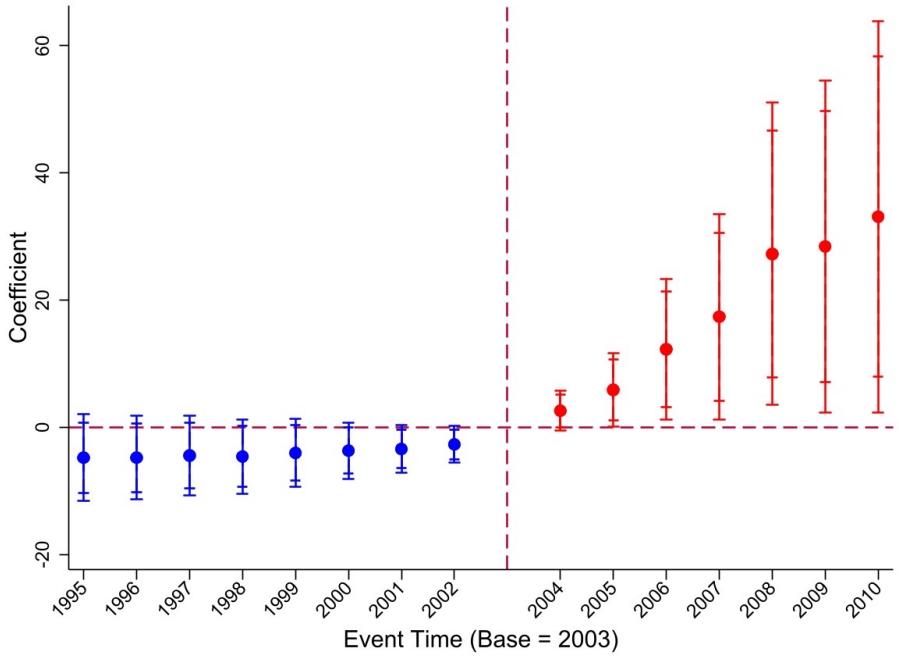
D Additional Tables and Figures

Figure A1: Border District: Rajasthan vs. Adjacent States

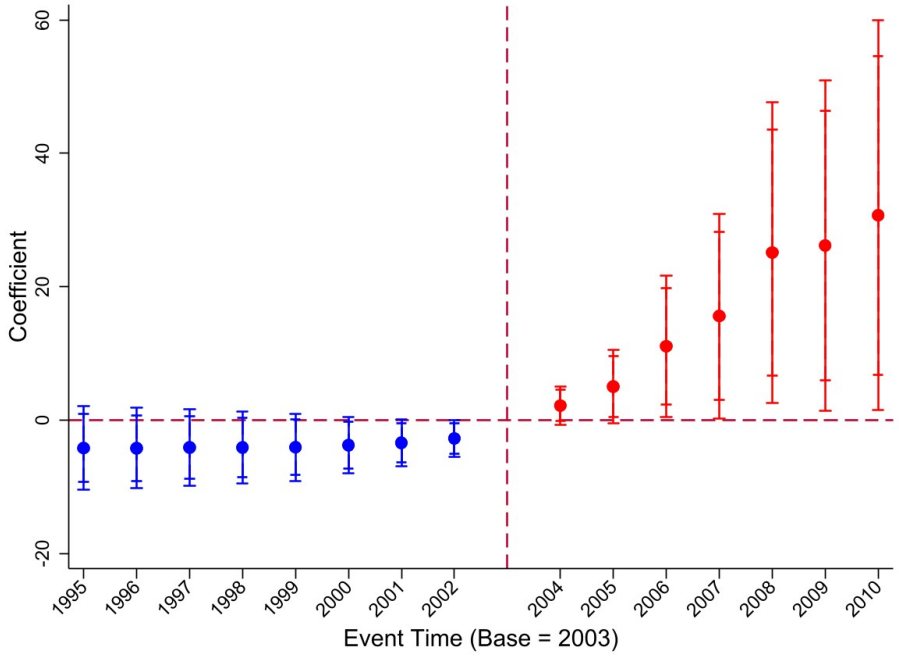


Notes: The figure shows the districts of Rajasthan (where there is a disproportionate expansion of colleges) and the districts of adjacent states used in the event-study estimation. Dark blue denotes Rajasthan's border districts and constitutes the treated group; light blue denotes Rajasthan's non-border districts. Dark green denotes the border districts of neighboring states and constitutes the control group; light green denotes the non-border districts of neighboring states. The neighboring states include Gujarat, Haryana, Madhya Pradesh, Punjab, and Uttar Pradesh.

Figure A2: College Expansion: Rajasthan vs. Adjacent States Border Districts



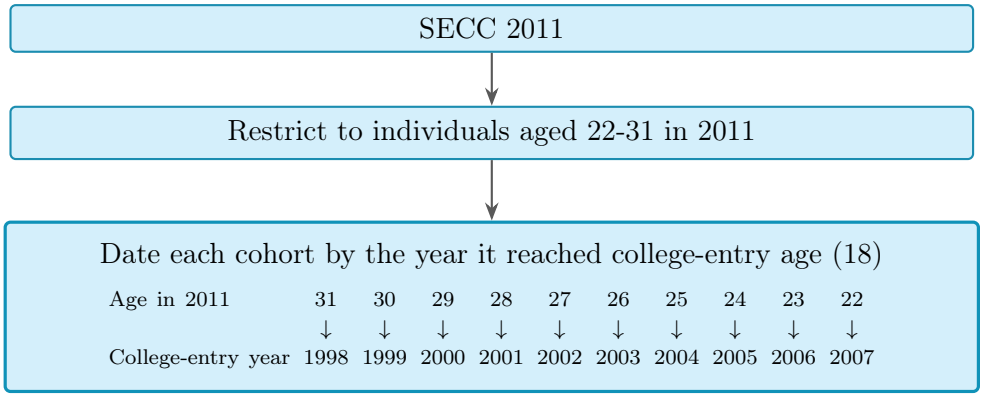
(a) Overall Colleges



(b) Private Colleges

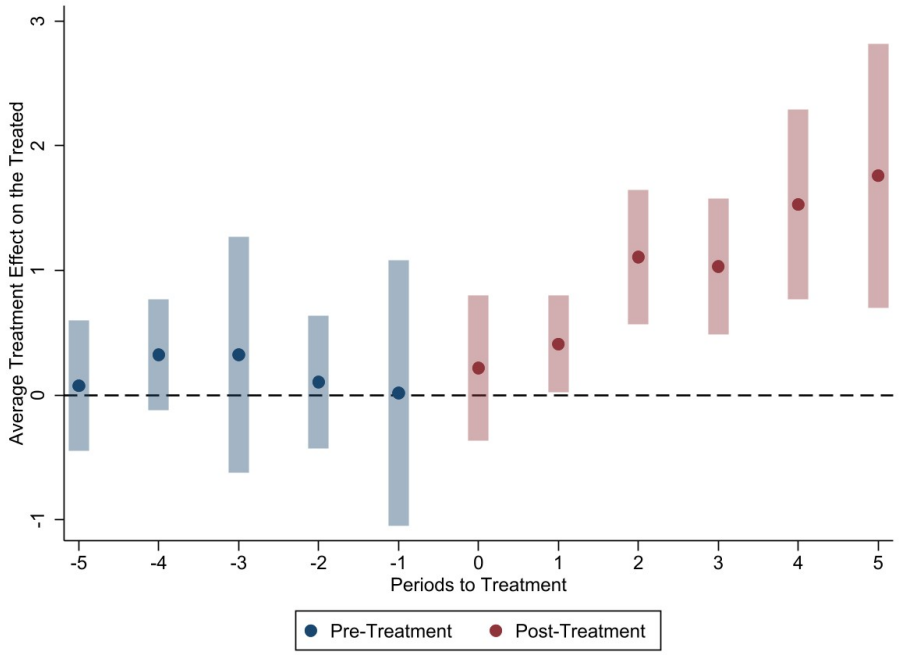
Notes: The figure plots coefficients from event study regressions estimating the differential trajectory of cumulative college counts in Rajasthan districts relative to border districts of neighboring states, using 2003 as the base year. The sample is restricted to border districts across six states—Gujarat, Haryana, Madhya Pradesh, Punjab, Rajasthan, and Uttar Pradesh—matched using 2001 Census district codes. The outcome variable is the cumulative number of colleges established in a district up to a given year. Panel (a) includes all colleges; panel (b) restricts to privately managed colleges. The outer caps denote 95% confidence intervals, and the inner caps denote 90% confidence intervals.

Figure A3: Construction of College-Entry Cohorts from SECC 2011



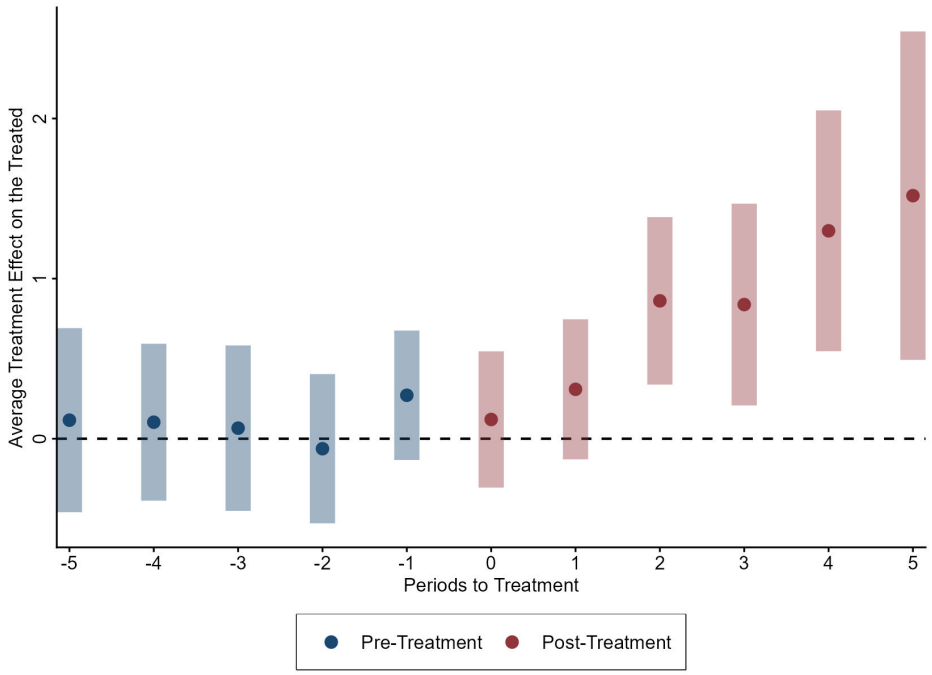
Notes: This figure outlines the construction of the village-cohort panel. From the SECC 2011, I restrict attention to individuals aged 22 to 31 in the census year. Each single-year age cohort is then assigned a calendar year corresponding to when it reached the college-entry age of 18, computed as 2011 − (age in 2011) + 18. This assigns the ten cohorts to college-entry years 1998 to 2007, allowing me to exploit variation in college access across these entry years.

Figure A4: CS-DiD Estimates at the 15 km Bandwidth, Controlling for Nightlights



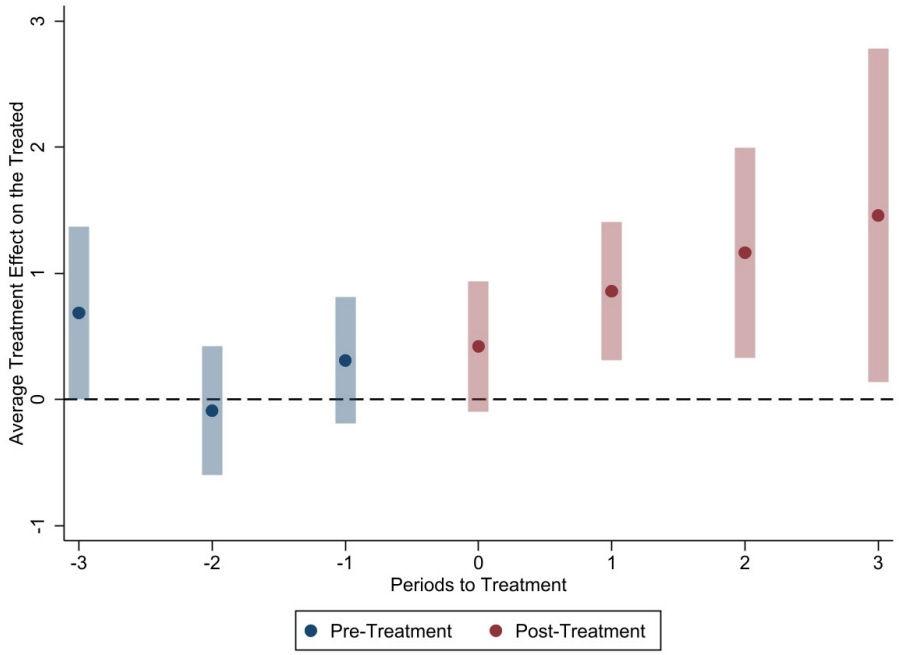
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. Estimates condition on subdistrict-level mean nighttime luminosity (DMSP) measured in the pre-treatment period $g - 1$, before any college opened nearby. ATT(g, t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure A5: CS-DiD Estimates at the 15 km Bandwidth, Bad-Control Adjustment



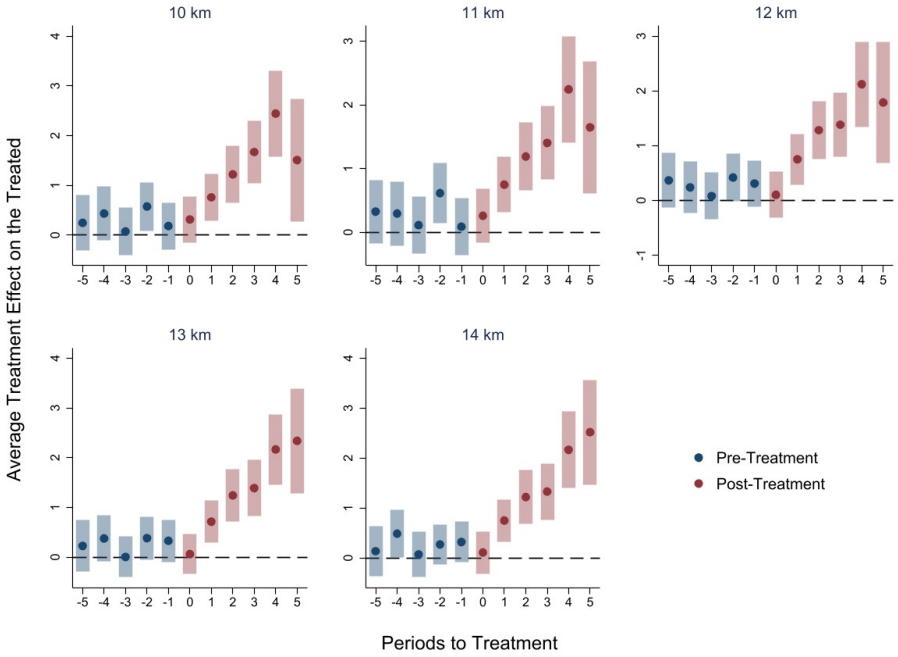
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. Subdistrict-level nighttime luminosity (DMSP) is treated as a bad control, since it may itself respond to college openings; its untreated evolution is modeled under covariate unconfoundedness, conditioning on its pre-treatment value and the lagged outcome. ATT(g,t) parameters are estimated using the parametric doubly-robust estimator, with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure A6: CS-DiD Estimates at the 15 km Bandwidth, College-Entry Cohorts 2000-2005



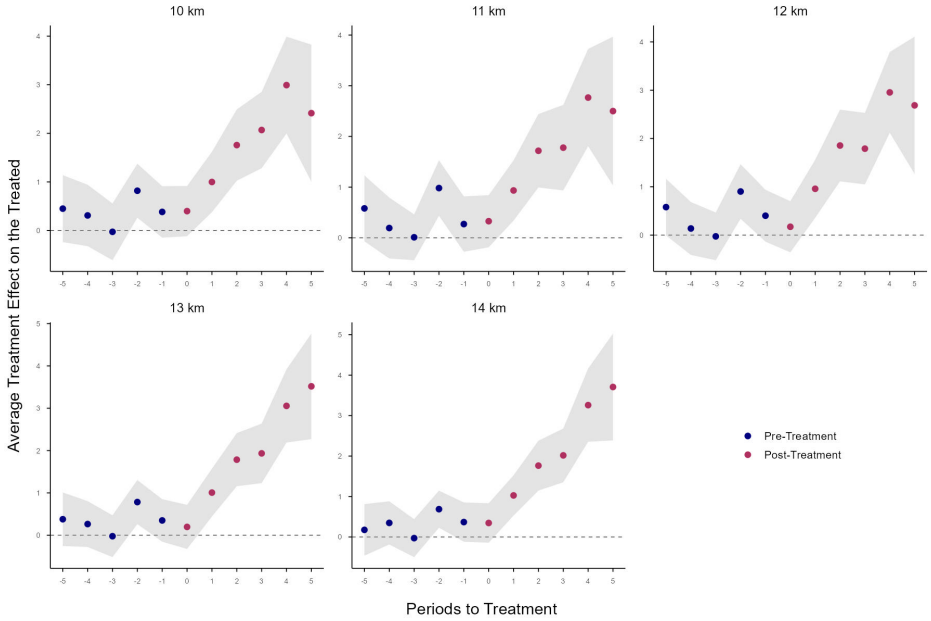
Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth and restricting the sample to cohorts that reached college-entry age between 2000 and 2005. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure A7: CS-DiD Estimates Across Bandwidths



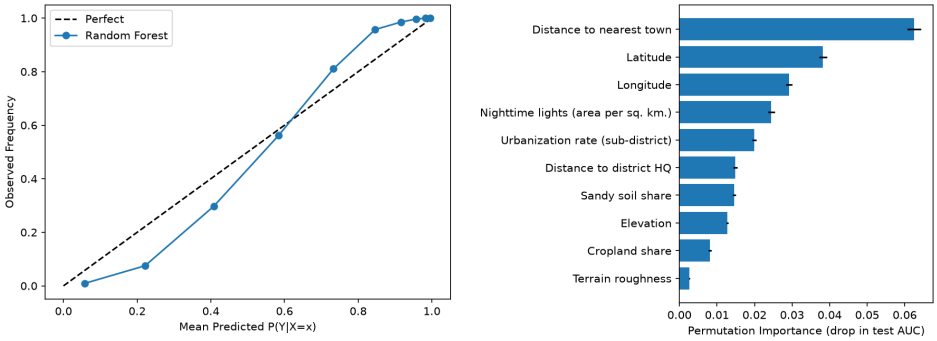
Notes: Each panel plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment for a given bandwidth, ranging from 10 to 14 km. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Figure A8: Continuous CS-DiD Estimates Across Bandwidths



Notes: Each panel plots average treatment-effect-on-the-treated (ATT) coefficients from continuous-treatment CS-DiD estimation against periods to treatment for a given bandwidth, ranging from 10 to 14 km. The outcome variable is the share of individuals completing tertiary education in each village-cohort cell. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator, with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

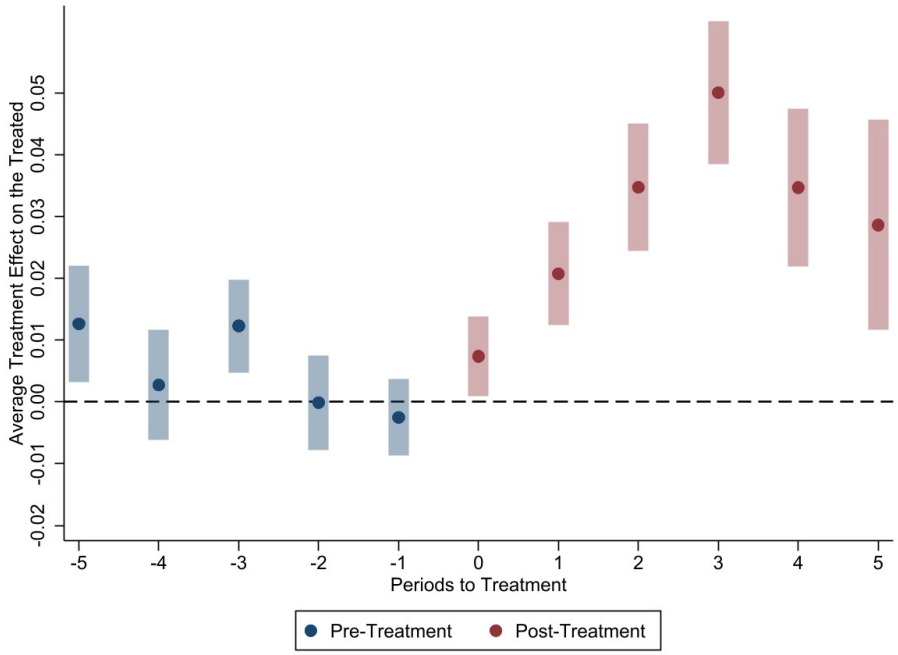
Figure A9: Village-Level Predictors of College Proximity



(a) Calibration (b) Top Predictors

Notes: A Random Forest classifier predicts whether a village lies within 15 km of a college established by 2011, using 31 geographic, terrain, demographic, land-use, and infrastructure predictors for 43,136 villages in Rajasthan. Panel (a) presents a reliability diagram based on a 25% hold-out sample, with villages grouped into ten equal-sized bins of predicted probabilities; the dashed line denotes perfect calibration. Panel (b) reports permutation importance for the top predictors.

Figure A10: CS-DiD Estimates at the 15 km Bandwidth, Night-Time Luminosity



Notes: The figure plots average treatment-effect-on-the-treated (ATT) coefficients from binary-treatment CS-DiD estimation against periods to treatment, using the 15 km bandwidth. The outcome variable is ln(1 + DMSP night-time luminosity) at the village level. ATT(g,t) parameters are estimated using the doubly-robust DiD estimator based on stabilized inverse-probability weighting and ordinary least squares (dripw), with not-yet-treated villages as controls. Blue markers denote pre-treatment periods and red markers denote post-treatment periods. Shaded bands denote 95% confidence intervals.

Table A1: Growth in Number of Colleges by State/Union Territory, 2003-2020

Rank	State/UT	N_{2003}	N_{2020}	Growth (%)	CAGR (%)
1	Dadra & Nagar Haveli and Daman & Diu	3	20	566.67	11.81
2	Rajasthan	552	3393	514.67	11.27
3	Haryana	241	1070	343.98	9.16
4	Uttar Pradesh	1649	7265	340.57	9.11
5	Arunachal Pradesh	10	41	310.00	8.65
6	Ladakh	2	7	250.00	7.65
7	Madhya Pradesh	854	2955	246.02	7.58
8	Gujarat	775	2421	212.39	6.93
9	Uttarakhand	136	421	209.56	6.87
10	Chhattisgarh	275	843	206.55	6.81
11	Tamil Nadu	942	2765	193.52	6.54
12	Himachal Pradesh	121	345	185.12	6.36
13	All India	15312	42857	179.89	6.24
14	Karnataka	1522	4057	166.56	5.94
15	Punjab	384	987	157.03	5.71
16	Puducherry	36	92	155.56	5.67
17	Jammu & Kashmir	132	335	153.79	5.63
18	West Bengal	560	1411	151.96	5.59
19	Andhra Pradesh	980	2448	149.80	5.53
20	Telangana	831	1987	139.11	5.26
21	Kerala	643	1477	129.70	5.01
22	Tripura	24	55	129.17	5.00
23	Jharkhand	168	364	116.67	4.65
24	Maharashtra	2181	4670	114.12	4.58
25	Sikkim	14	28	100.00	4.16
26	Bihar	586	1001	70.82	3.20
27	Nagaland	41	66	60.98	2.84
28	Goa	40	63	57.50	2.71
29	Odisha	823	1249	51.76	2.48
30	Andaman & Nicobar Islands	6	9	50.00	2.41
31	Meghalaya	49	73	48.98	2.37
32	Manipur	70	103	47.14	2.30
33	Mizoram	28	40	42.86	2.12
34	Delhi	149	188	26.17	1.38
35	Assam	463	581	25.49	1.34
36	Chandigarh	22	24	9.09	0.51
37	Lakshadweep	0	3	—	

Notes: Source - AISHE, Ministry of Education, Government of India, 2021-22. N_{2003} and N_{2020} denote the number of colleges in 2003 and 2020, respectively. Growth (%) is the cumulative percentage change in the number of colleges between 2003 and 2020, defined as $\left(\frac{N_{2020} - N_{2003}}{N_{2003}}\right) \times 100$. CAGR is the compound annual growth rate over the same 17-year period, defined as $\left(\frac{N_{2020}}{N_{2003}}\right)^{1/17} - 1$, expressed as a percentage. States and UTs are ranked in descending order of CAGR. Growth (%) and CAGR are undefined for Lakshadweep because the number of colleges in 2003 is zero.